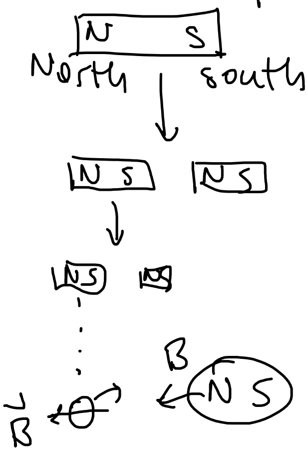
Magnetic Fields

Friday, April 26, 2024 3:14 PM

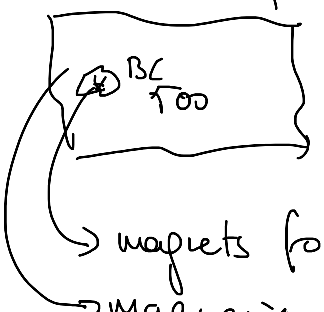
EM force

↳ magnetic = magnets



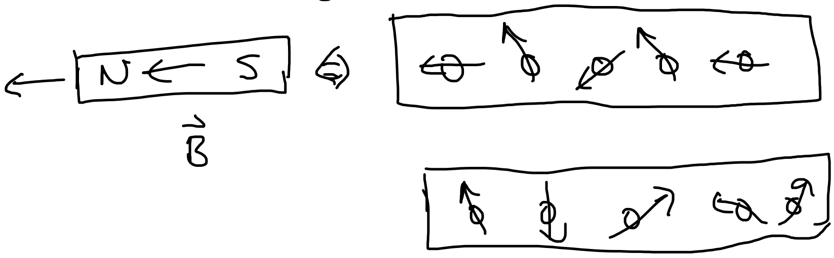
single atom } "quantum"

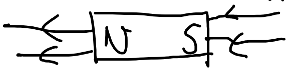
Türkey

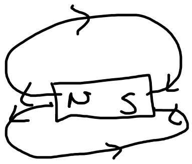


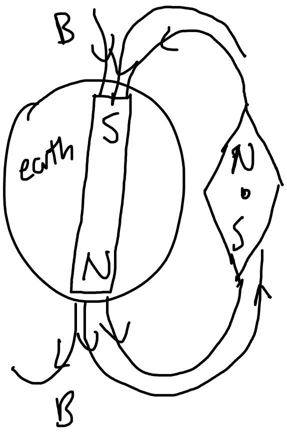
↳ magnets found (lithos)

↳ Magnesia





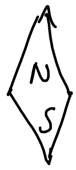




Compass = pusula

$\vec B_{earth} = 1$ gauss

$= 10^{-4}$ Tesla



$[\vec B]$ = Tesla

earth 10^{-4} T

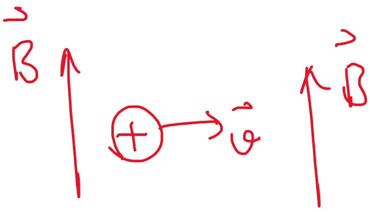
fridge magnets $10^{-3} - 10^{-2}$ T

MRI (emar) NMR 1–2 T

research labs → 45 T

1 Tesla is a HUGE magnetic field.

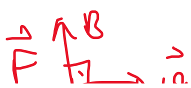
Force on a charged particle in a magnetic field



$\vec F = q\,\vec v \times \vec B$

↳ vector product

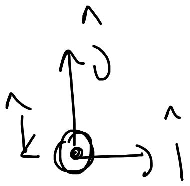
$\vec v \times \vec B = |\vec v|\,|\vec B| \sin\theta$



$\vec v \times \vec B = ?$

$\odot$ out of page

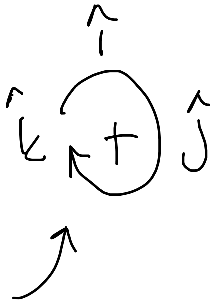
$|F| = |q v B \sin\theta|$



$\hat{i}\times\hat{j} = \hat{k}$

$\hat{j}\times\hat{k} = \hat{i}$

$\hat{k}\times\hat{i} = \hat{j}$



$\hat{i}\times\hat{k} = -\hat{j}$

$\hat{k}\times\hat{j} = -\hat{i}$

$\hat{j}\times\hat{i} = -\hat{k}$

$\vec{F} = q\vec{v}\times\vec{B}$

$[N] = [C \ \frac{m}{s} \ T]$

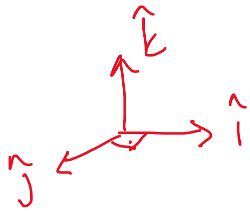
$[1 \text{ Tesla} = \frac{Ns}{mC} = \frac{N}{m\frac{C}{s}} = \frac{N}{mA}]$

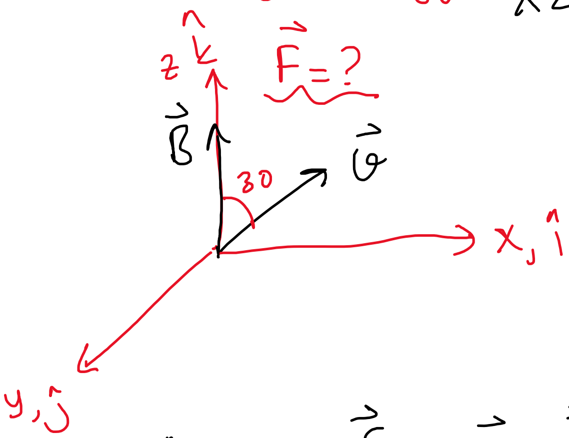
$[\frac{kg}{s^2A}] = \frac{kg \, m/s^2}{mA}$

ex) $q = +1.6\times10^{-19}C$ $\qquad \vec{B} = 2T\hat{k} = B\hat{k}$

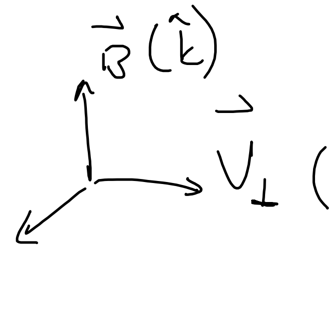
$|\vec{v}| = 3\times10^5 \, m/s$ $\qquad \vec{v}$ is 30° with z axis.

$\vec{v}$ is on xz plane.





$\vec{F} = ?$



$\vec{v} = v\sin 30\hat{i} + v\cos\theta\hat{k}$

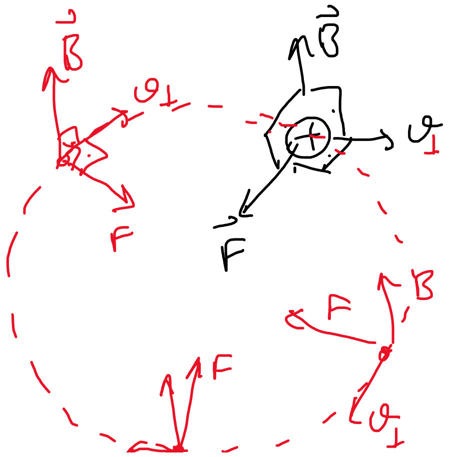
$\vec{F} = q\vec{v}\times\vec{B} = q(v\sin30\hat{i} + v\cos3$

$= qvB(\sin30 \underbrace{\hat{i}\times\hat{k}}_{\hat{j}} + \cos3$

$\vec{F} = qvB\sin30\hat{j}$

$= 1.6\times10^{-19}\times3\times10^5\times2\times\frac{1}{2}$

$\boxed{\vec{F} = 4.8\times10^{-14} N\hat{j}}$



$F = qvB\sin\theta$

$\vec{v}_\perp$

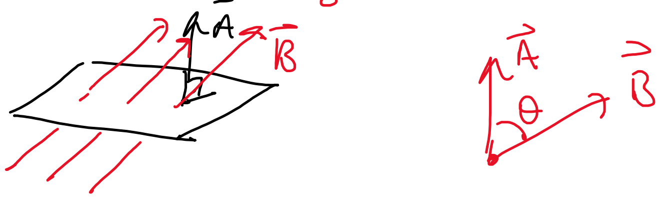
$$\vec{v}\times\vec{B} = \left(\vec{v}_{\parallel} + \vec{v}_{\perp}\right)\times\vec{B}$$
$$= \underbrace{\vec{v}_{\parallel}\times\vec{B}}_{\text{zero}} + \underbrace{\vec{v}_{\perp}\times\vec{B}}_{\neq 0}$$

electric flux $\Phi_E = \vec{E}\cdot\vec{A}$

↳ gauss law

magnetic flux $\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta$

$[\Phi] = \text{weber} = \text{Wb}$



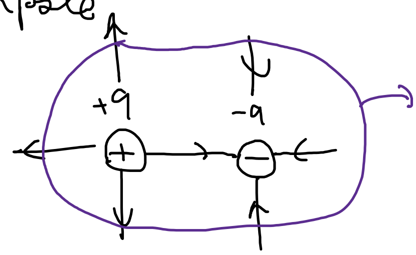
Gauss Law $\Phi_E = \oint \vec{E}\cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$ (1st maxwell equ)

gauss law for magnetism

$$\boxed{\Phi_B = \oint \vec{B}\cdot d\vec{A} = 0}$$ 2nd maxwell e

there is no magnetic monopole "for now"

dipole



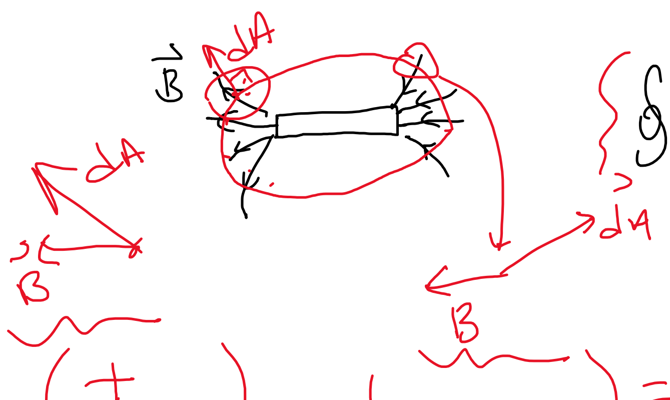
$\Phi_E = 0 = \frac{q_{in}}{\varepsilon_0} = 0$

$\oplus\ \ominus$ = N S

dipole

$\Phi_B = 0 = \oint \vec{B}$

no monopole!!

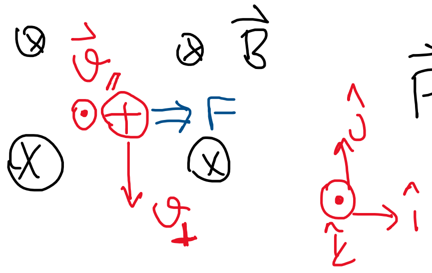


$\left\{ \oint \vec{B}\cdot d\vec{A} = \Phi_B = 0 \right\}$

↳ No magnetic monopole

$\underbrace{(+)}\ \underbrace{(-)} = 0$

Force due to $\vec{R}$



$$\vec{F} = q\,\vec{v} \times \vec{B}$$

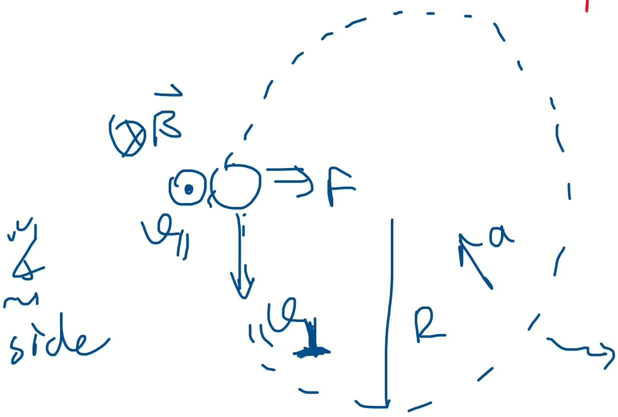
$v \times B = (v_\parallel + v_\perp) \times B$

$\perp$ to both vectors.

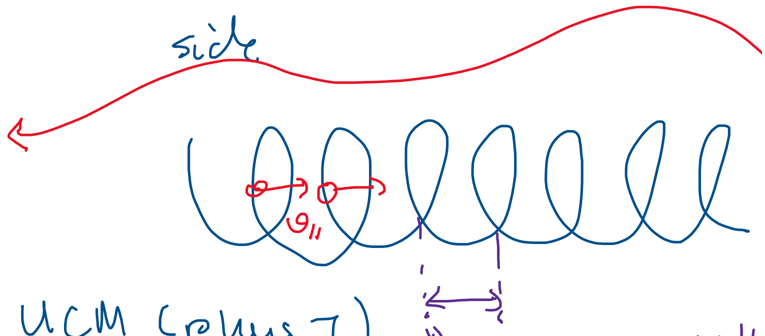
$\Rightarrow \vec{v}_\parallel \times \vec{B} = 0$

$\vec{v}_\perp \times \vec{B} = v_\perp B s$

$$\vec{F} = q v_\perp B$$



UCM (phys I)



"pitch length"

adım uzunluğu

$F = ma_r$

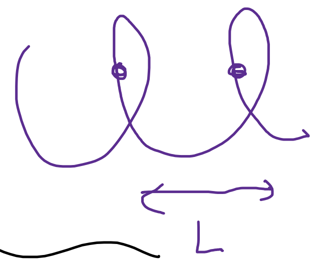
$a = \frac{v_\perp^2}{R}$

$q v_\perp B = m \frac{v_\perp^2}{R} \Rightarrow v_\perp = \frac{qBR}{m} = \text{const} = \frac{\Delta x}{\Delta t} = \frac{2\pi R}{T}$

$$v_\perp = \frac{qBR}{m}$$

q, B, m const $\quad v_\perp \uparrow \quad R \uparrow$

$\frac{qBR}{m} = \frac{2\pi R}{T}$



$L = v_\parallel T$

$T = \frac{2\pi m}{qB}$

ex) $q = 1.6 \times 10^{-19}$ C

$m = 1.67 \times 10^{-27}$ kg

$\hat{i} = B_x = 0.5$ T

$v_\parallel \; \hat{i} = v_x = 1.5 \times 10^5$ m/s

$v_y = 0$

$v_\perp \; \hat{k} = v_z = 2 \times 10^5$ m/s

$\vec{F} = ? \quad a = ? \quad R = ?$

$L = \text{pitch length} = ?$

$\vec{F} = q\,\vec{v} \times \vec{B}$

$\vec{F} = q(\vec{v}_\parallel + \vec{v}_\perp) \times \vec{B}$

$v_\perp = 2 \times 10^5 \hat{k} \times 0.5\text{T}\, \hat{i}$

$\hat{k} \times \hat{i} = +\hat{j}$

$F = q v_\perp B = 1.6 \times 10^{-19} \times 2 \times 10^5 \times$

$\vec{F} = 1.6 \times 10^{-14}\,\text{N}\,\hat{j}$

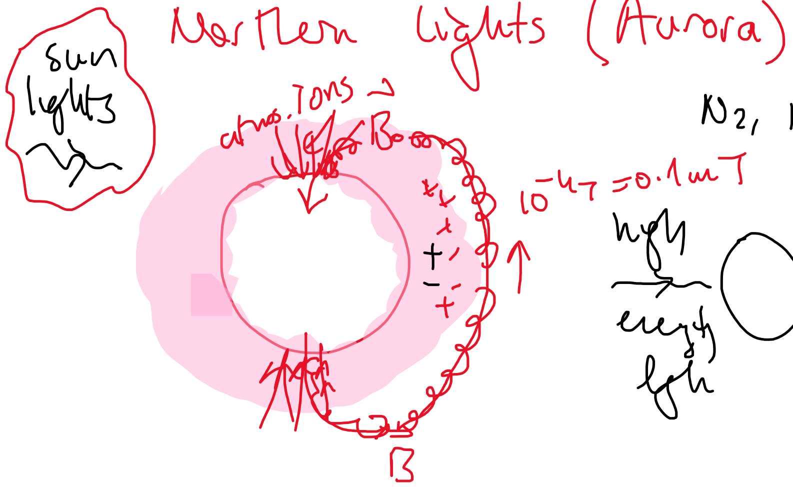
$a = \frac{F}{m} = \frac{1.6 \times 10^{-14}}{1.67 \times 10^{-27}} = 9.$

$R = ?$ $\quad F = \frac{m v_\perp^2}{R} = q v_\perp B$

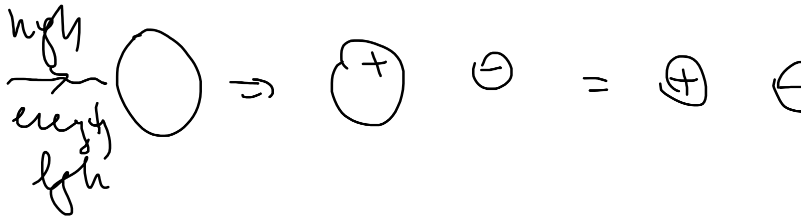
$$R = \frac{m v_\perp}{qB} = \frac{1.67\times10^{-27} \times 2\times10^5}{1.6\times10^{-19} \times 0.5} = 4.2\times10^{-3}\,m = 4.2\,mm$$

$$L = v_\parallel T = v_\parallel \frac{2\pi R}{v_\perp} = 2\times3.14\times 4.2\times10^{-3} \frac{1.5\times10^5}{2\times10^5} = \underline{\underline{19.7\,m}}$$

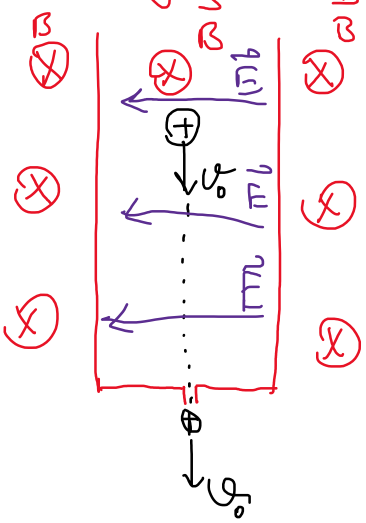
Northern Lights (Aurora)

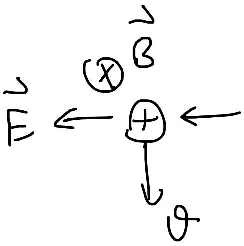


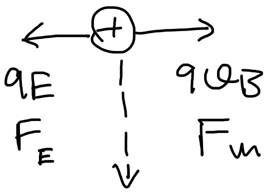
$N_2, H_2, O_2, CO_2 \ldots$



Velocity Selector. (mass spectrometer)



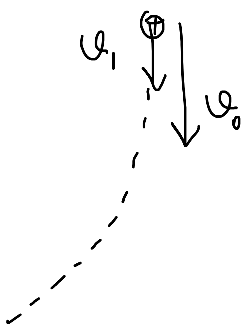




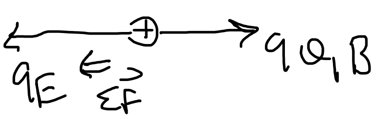
$F_m = q\vec{v}\times\vec{B}$ $\qquad \Sigma F = 0$

$qE = qvB$

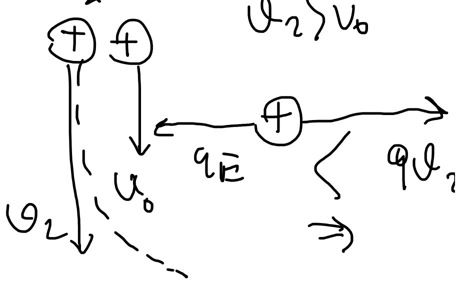
$$\boxed{v_0 = \frac{E}{B}}$$



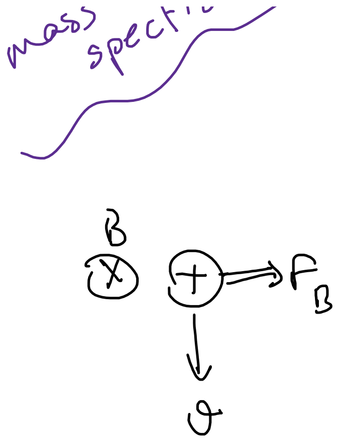
$v_1 < v_0$



v_2 $\qquad v_2 > v_0$



meter.



mass spectrometer measures mass of atoms

$C^{12}H_1 =$

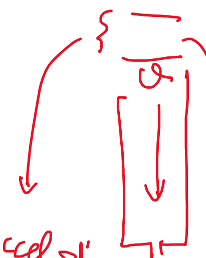
$CH_4 = 1$

measure R, m ✓

$$q v B = \frac{m_1 v^2}{R_1} \qquad \uparrow \frac{m_1}{\uparrow R_1} = \frac{qB}{v_0}$$

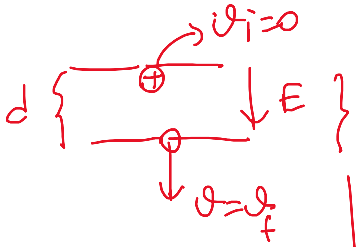
$$\frac{m_2}{R_2} = \frac{m_1}{R_1} = \frac{qB}{v}$$

Thomson's $\frac{e}{m}$ ratio expt. 1897 (Nobel)



accelerating the particle

$$qE \leftarrow \oplus \rightarrow qvB \qquad \boxed{v = \frac{E}{B}}$$



$$q\Delta V = U = \frac{1}{2} m v^2$$

$$\boxed{q\Delta V = \frac{1}{2} m \frac{E^2}{B^2}}$$

$$\frac{q}{m} = \frac{E^2}{2\Delta V B^2} = 1.7588\times10^{11}\ \frac{C}{kg}$$

$$\frac{e}{m} =$$

1912 Millikan $e = 1.6\times10^{-19}\,C$ ✓ $\Rightarrow m_e =$ ✓

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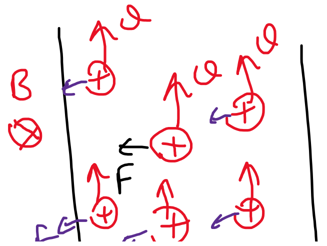
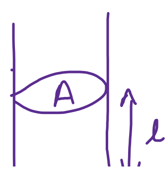
magnetic force on current carrying wire

v = drift velocity.



Vol = $A\ell$



$\sum \vec F_q$ = force acting on wire

n = charge density = $\frac{\text{number}}{\text{Vol}}$

in the wire

$F_q = q v B$

no of charge $=$ n(Volume)

(no of charge) $F_q = \Sigma F_q$

$n \underbrace{vol.}_{Al} q v B$

$\Sigma F_q = n A l q v B$

↳ drift vel.

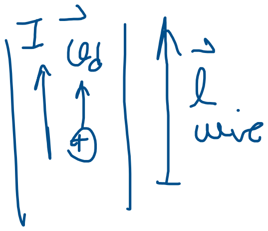
prev. chapter

$I = nqAv_d$

$\vec F = q\vec v \times \vec B$

$F = qvB$

$\Sigma \vec F_q = \underbrace{nAq\vec v}_{I}\, l\, B$



$$\boxed{\vec F_{wire} = I\, \vec l \times \vec B}$$

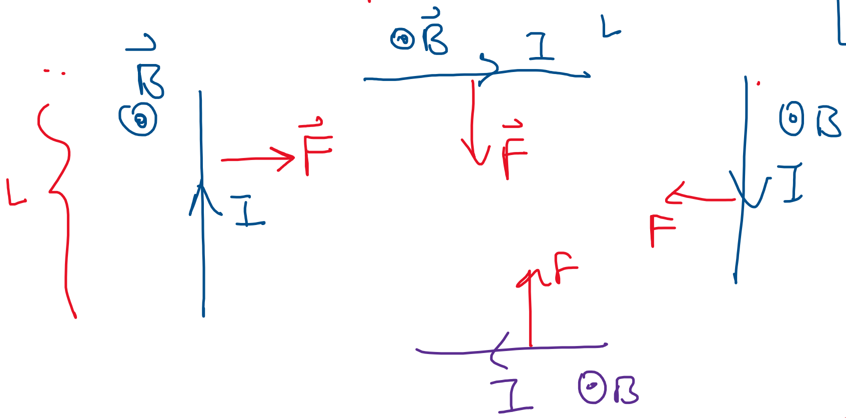
$$\boxed{\vec F = q\vec v \times \vec B}$$

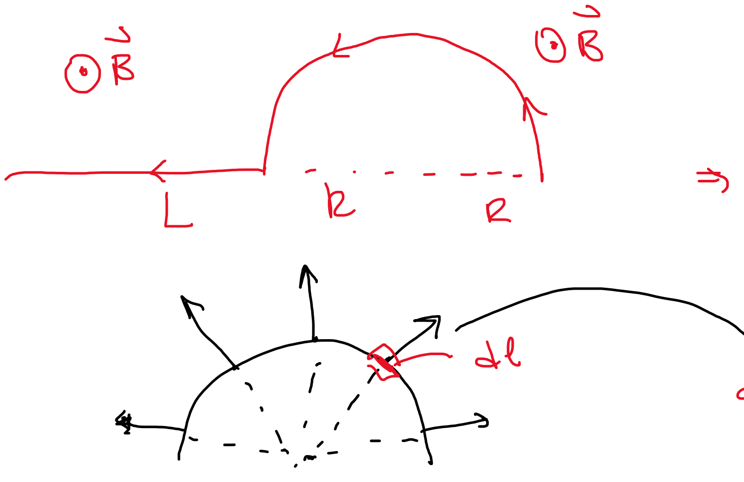
$\vec J = \frac{I}{\vec{Area}}$

$[N = m\, A\, T]$

$$\boxed{T = \frac{N}{mA}}$$

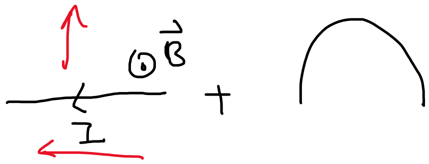
$\vec F = I\vec l \times \vec B$

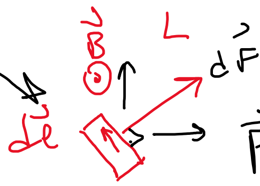




$\Rightarrow$

$F_I = ILB\sin 90$

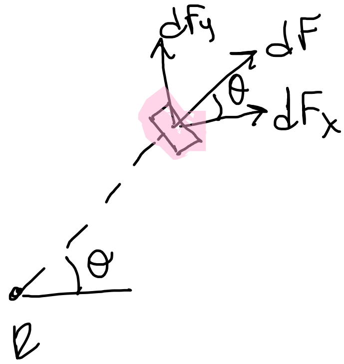




$\vec F = I\vec l \times \vec B$

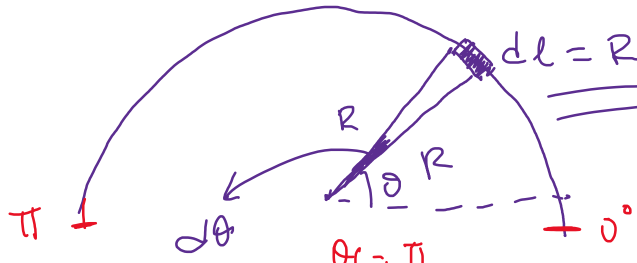
$d\vec F = I\, d\vec l \times \vec B$

$dF = I\,dl\,B$



$I\,dl\,B$

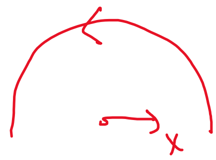
$dF_x = dF\cos\theta \qquad dF_y = dF\sin\theta$



$dl = R\,d\theta$

$$\int dF_x = \int I\,dl\,B\cos\theta = \int_{\theta_i=0}^{\theta_f=\pi} I\,R\,B\cos\theta\,d\theta = IRB\,[\sin\theta]$$

$$F_x = IRB\,[\sin\pi - \sin 0] = 0$$



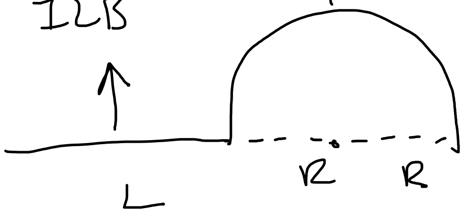
$F_x = 0$

$$\int dF_y = I\,dl\,B\sin\theta = \int_0^{\pi} I\,R\,B\,\sin\theta\,d\theta = IRB\,[-\cos\theta]_0^{\pi}$$

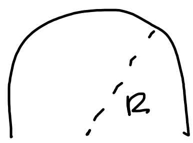
$$F_y = IRB\,[\underbrace{-\cos(\pi)}_{-(-1)} + \overbrace{\cos(0)}^{1}] = 2IRB$$

$\uparrow 2IRB = I\,2R\,B$

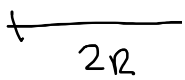
ILB

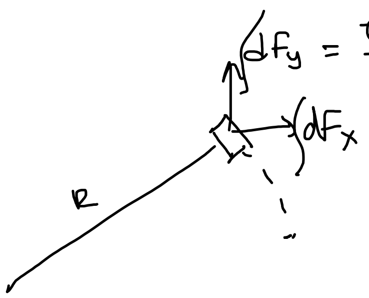


$\Rightarrow$



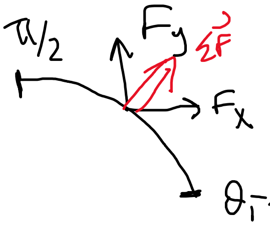
$\equiv$





$$\int dF_y = IRB\,[-\cos\theta]_{\theta_1}^{\theta_2}$$

$$\int dF_x = IRB\,[\sin\theta]_{\theta_1}^{\theta_2}$$

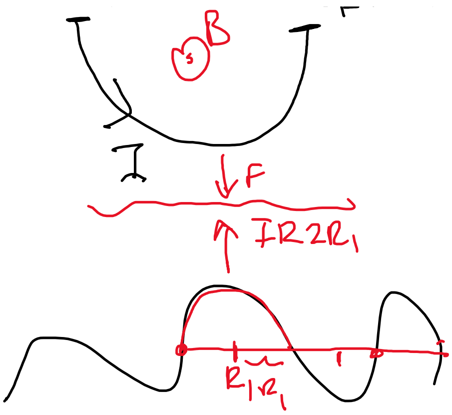


$\theta_f = \pi/2 \qquad \theta_i = 0$

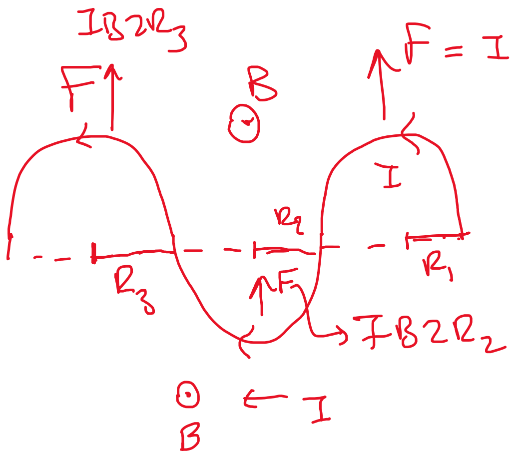
$$F_y = IRB\left[-\cos\frac{\pi}{2} + \cos 0\right] = IRB \qquad \left|\sum\vec{F}\right| = \sqrt{2}\,IRB$$

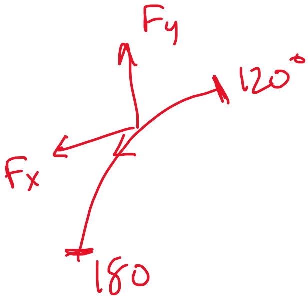
$$F_x = IRB\left[\sin\frac{\pi}{2} - \sin 0\right] = IRB$$

$\theta_i = \pi \qquad \theta_f = 2\pi$



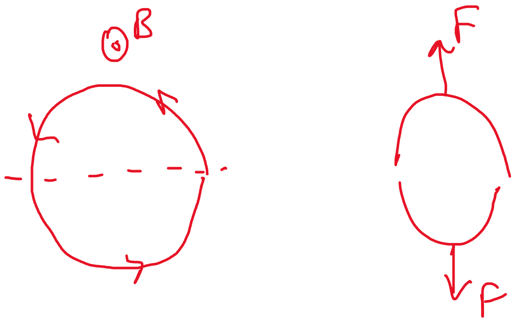
$$\int dF_x = 0 \qquad F_y = 2IRB$$





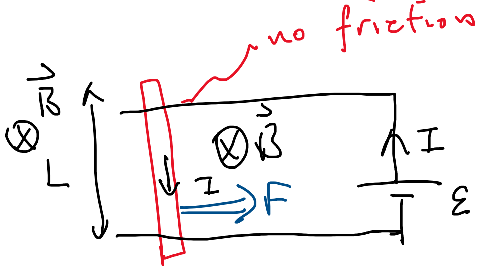
$$\left[-\cos\theta\right]_{120}^{180}$$

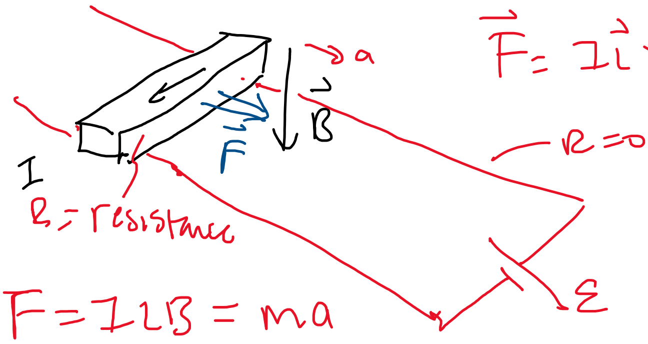
$$\left[\sin\theta\right]_{120}^{180}$$



$$\left[-\cos\theta\right]_0^{2\pi} = 0 = \left[\sin\theta\right]_0^{2\pi}$$

$$\Sigma F = 0$$





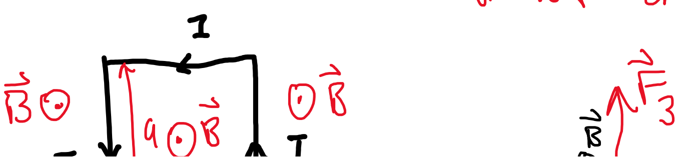
$$\vec{F} = I\vec{L}\times\vec{B}$$

$$F = ILB = ma$$

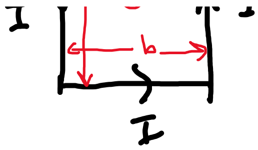
$$I = \frac{\varepsilon}{R}$$

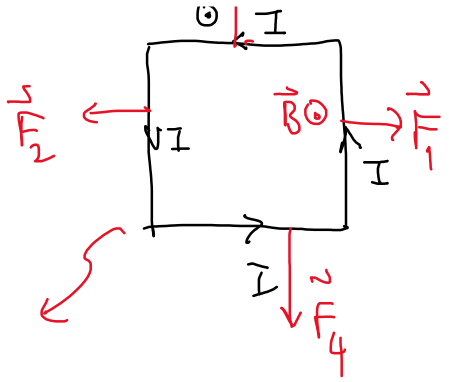
$$a = \frac{ILB}{m} = \frac{\varepsilon LB}{Rm}$$

FORCE on a CURRENT CARRYING LOOP



$$\vec{F} = I\vec{L}\times\vec{B}$$



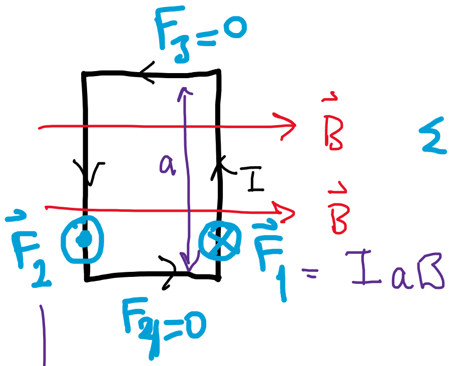


$$\int d\vec F = \int I d\vec l \times \vec B$$

$$\sum \vec F = 0$$

$$F_1 = IaB = F_2$$

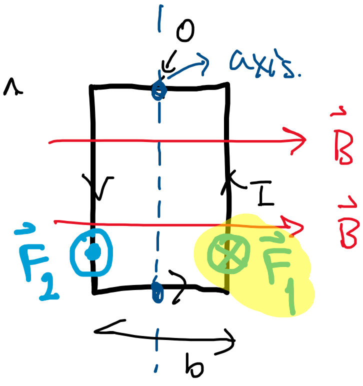
$$F_3 = F_4 = IbB$$

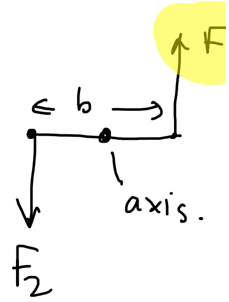


$\vec F_1 = IaB$

IaB

$$\sum \vec F = 0$$





Rotates

$$\sum \tau = F_2 \frac{b}{2} + F_1 \frac{b}{2}$$

$$= IaB\frac{b}{2} + IaB\frac{b}{2}$$

ROTATE

torque

$$\tau = F_2 \frac{b}{2} + F_1$$

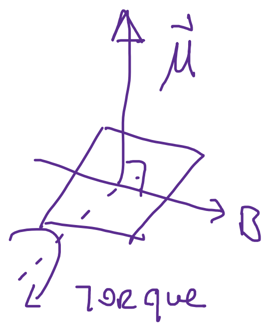
$$\sum \tau = I\underbrace{ab}_{\text{area of loop}}B \quad (\text{torque}) = IAB$$

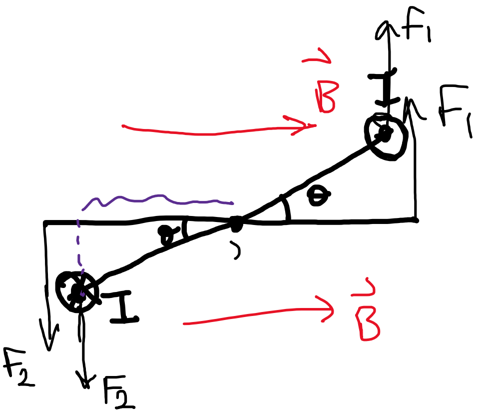
$$\sum \tau = I\underbrace{A}B$$

$\tau\uparrow$; $I\uparrow$; $A\uparrow$, $B\uparrow$

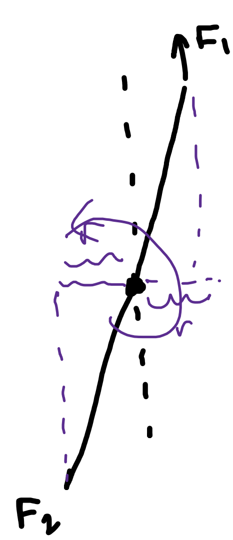
$\mu \equiv$ magnetic dipole moment

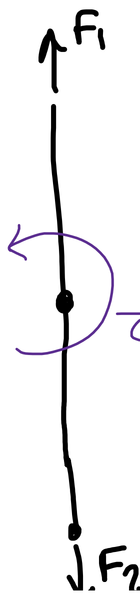
$$\vec\tau = \vec\mu \times \vec B$$





$$I\vec L \times \vec B = \vec F$$

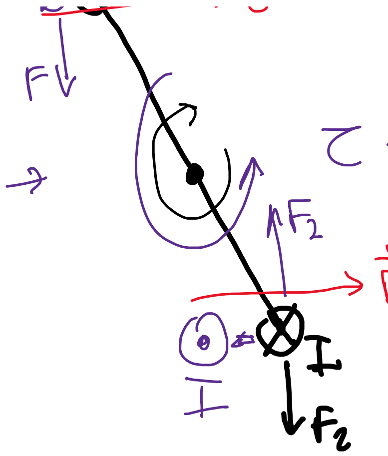




$$\vec r \times \vec F = 0$$

$$\tau = 0$$

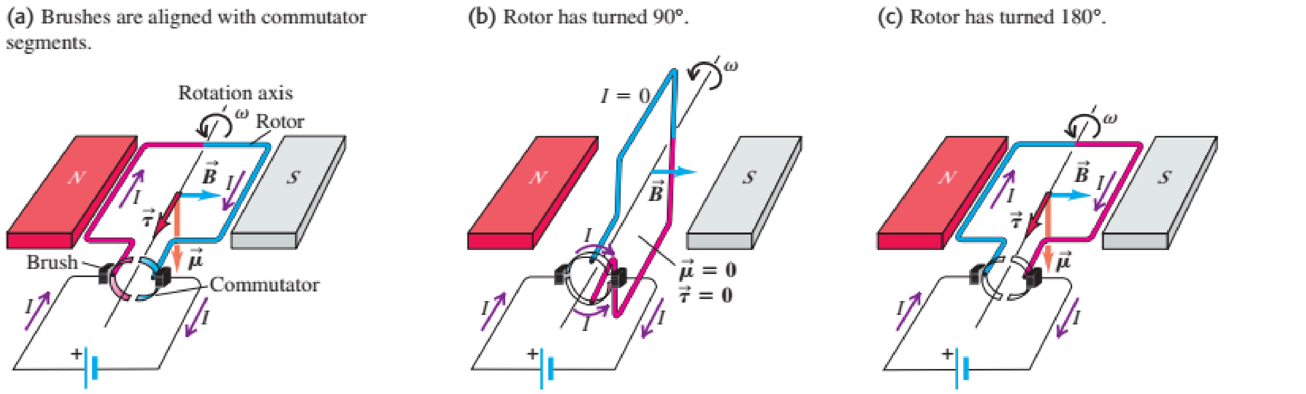
$$\vec\tau = \vec r \times \vec F$$



τ = other direction

* if we want keep rotating the loop (torque in same direction always flip I direction.

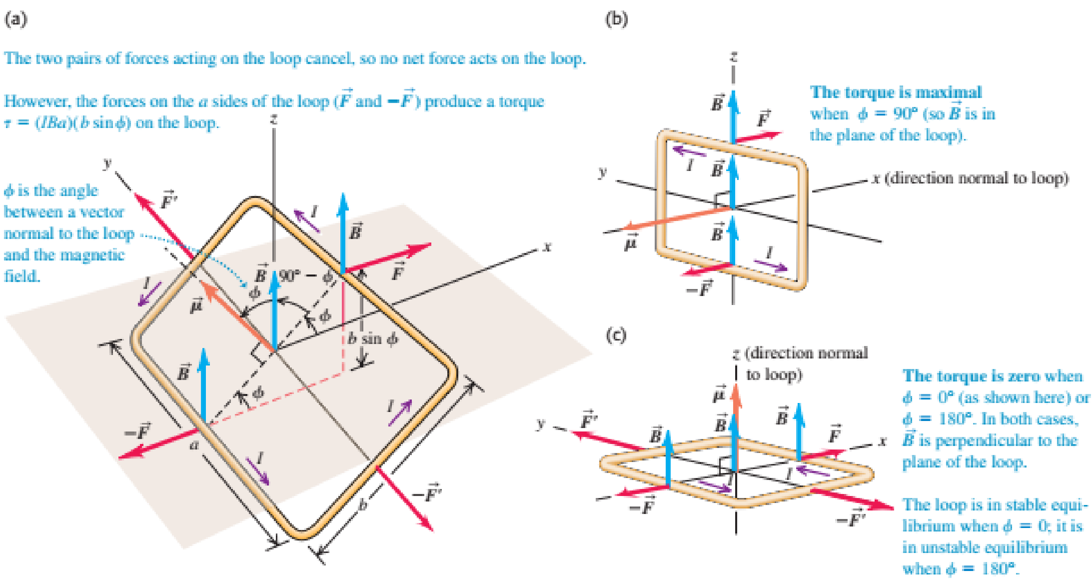
DC Motor

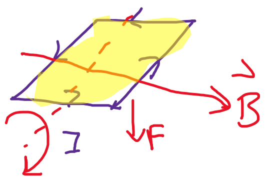


(a)
- Current flows into the red side of the rotor and out of the blue side.
- Therefore the magnetic torque causes the rotor to spin counterclockwise.

(b)
- Each brush is in contact with both commutator segments, so the current bypasses the rotor altogether.
- No magnetic torque acts on the rotor.

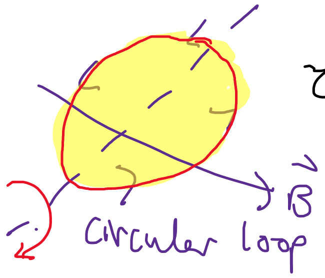
(c)
- The brushes are again aligned with commutator segments. This time the current flows into the blue side of the rotor and out of the red side.
- Therefore the magnetic torque again causes the rotor to spin counterclockwise.





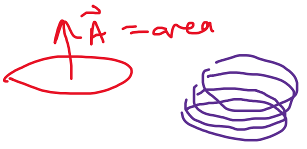
$$\tau = \underbrace{IA}_{\mu}B \quad A \rightarrow \text{area}$$

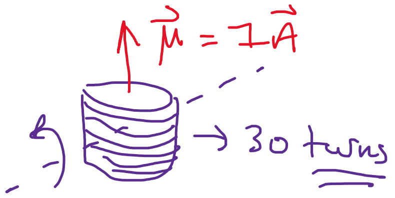
I ⊗ → $\vec{B}$ ↓ F



$\tau = IAB$ ↓ area of circle.

ex) a circular coil has 0.05m radius, it has 30 turns.





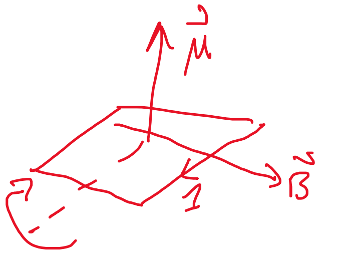
area $= \pi r^2 = 3.14(0.05)^2 = A$

$B = 1.2\,T$ $\quad I = 5A$

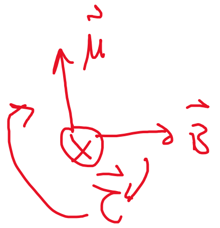
$\tau = ?$

$\tau = I A B N$ (# of turns, $N = 30$)

$\tau = 1.41\ Nm$

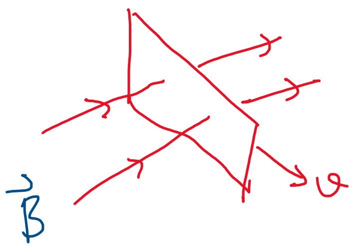


$\tau = \vec{\mu} \times \vec{B}$

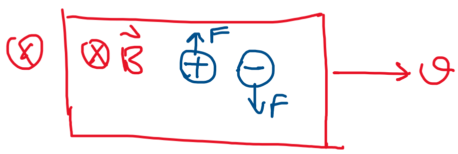


HALL EFFECT

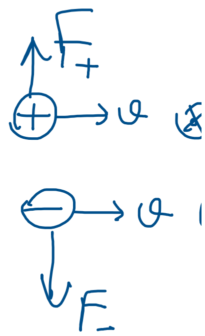
metal is placed and moven in a $\vec{B}$ field
(dragging)

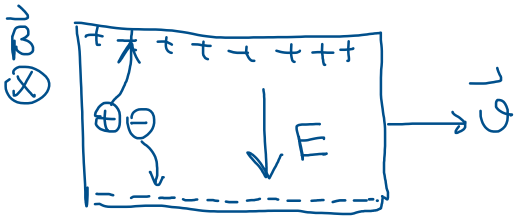


$\vec{F} = q\vec{v} \times \vec{B}$

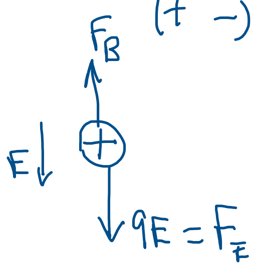


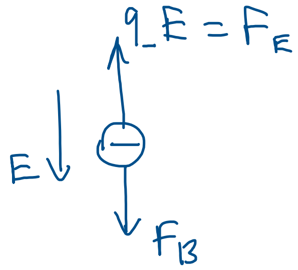
$q_- < 0$





$F = q\,\underbrace{\vec{v} \times \vec{B}}_{up}$
$(+\ -)$

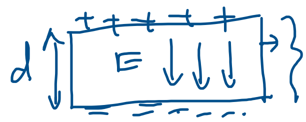




$F_B = F_E$
⊕, ⊖ charges will stop going towards edges/ sides.

$qvB = qE$

$\underbrace{vB} = E = \frac{\Delta V}{d}$



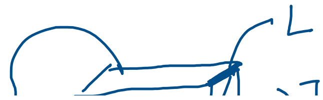
ΔV = pot. difference.

$B = \frac{E}{v} = \frac{\Delta V}{d}\frac{1}{v_d}$

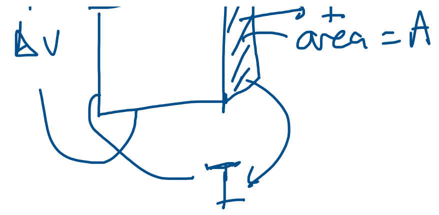
v = drift velocity $= v_d$

$I = nqAv_d \Rightarrow v_d = \underbrace{\frac{I}{nqA}}$

$\ldots\ nqA = dL$



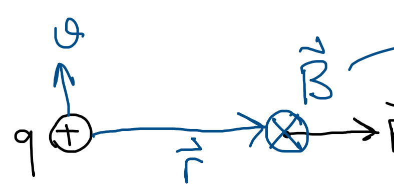
$B = \frac{\Delta V \cdots}{d\, I}$



— End Ch 27 — magnetic force on $\oplus \to v$, $\vec{I}$

Sources of magnetic field (How to produce $\vec{B}$)

"charge in motion produce magnetic field"



$$\vec{B} = \frac{\mu_0}{4\pi}\, q\, \frac{\vec{v} \times \hat{r}}{r^2}$$

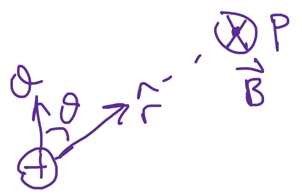
$\vec{E} = \frac{q}{4\pi\varepsilon_0} \frac{1}{r^2} \hat{r}$

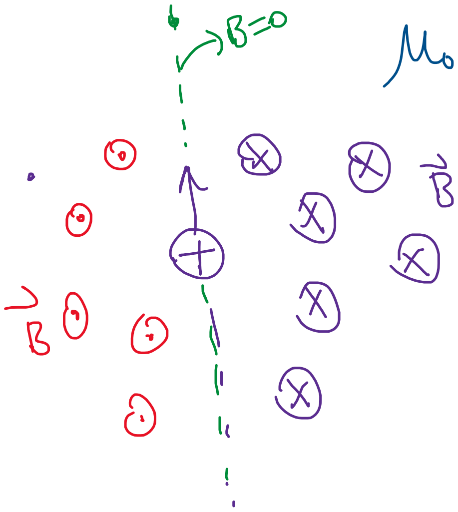
$\hat{r} = \frac{\vec{r}}{|\vec{r}|}$ = unit vector.

$\vec{v} \times \hat{r} = |\vec{v}|\, |\hat{r}| \sin\theta$

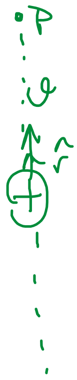
μ_0 = magnetic permitivity of space

$\mu_0 = 4\pi \times 10^{-7}\ \frac{Tm}{A}$





$\vec{v} \times \hat{r}$



$\vec{v} \times \hat{r} = 0$

$|\vec{v}\, \hat{r}| \sin\theta$

μ_0 = 'mu'-zero =

$\frac{1}{\sqrt{\mu_0 \varepsilon_0}} = C$ = speed of light

ε_0 = electric permitivity of space

$\varepsilon_0 = 8.82 \times 10^{-12}$ (- - -)

$\mu_0 = 4\pi \times 10^{-7}\ \frac{Tm}{A}$

$\frac{\mu_0}{4\pi} = 10^{-7}$

$\frac{\mu_0}{4\pi} = k'$

$\mu_0 = k'\, 4\pi$

$B = \frac{\mu_0}{4\pi}\, q \cdot \frac{\vec{v} \times \hat{r}}{r^2}$

$E = \frac{kq}{r^2}$ $\quad k = \frac{1}{4\pi\varepsilon_0}$

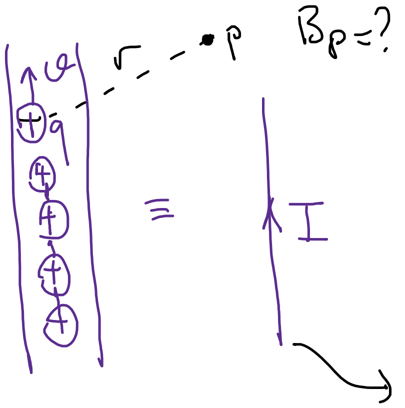
$\mu_0 \sim \frac{B r^2}{q v \hat{r}} = \frac{Tm^2}{C\, m}$

$\hat{r}$ → unitless

$\varepsilon_0 = \frac{1}{k\,4\pi}$

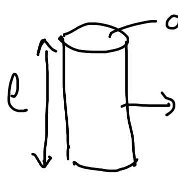
$\mu_0 = \frac{Tm}{\frac{C}{s} = \text{ampere}} = \left[\frac{Tm}{A}\right]$

$B_p = ?$



$\Sigma B = B$ created by I current carrying wire

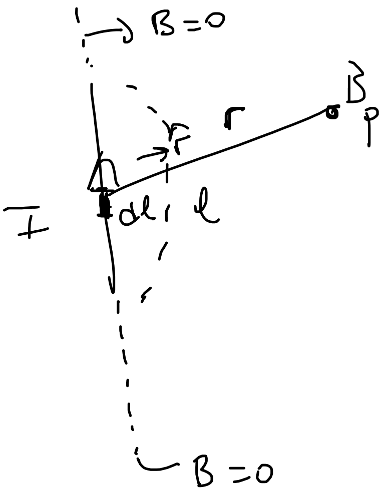
$$\vec{B} = \frac{\mu_0}{4\pi} q \frac{\vec{v} \times \hat{r}}{r^2}$$

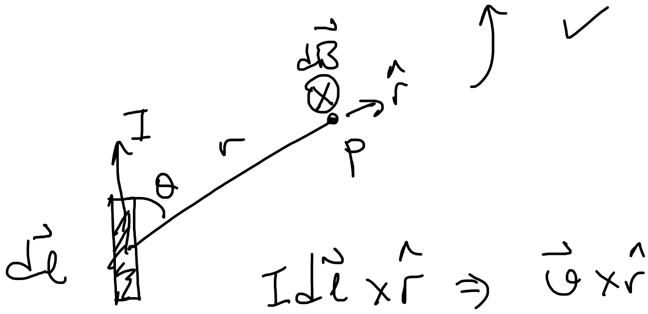


# of charges within this volume $= n$ Volume $= n A l$

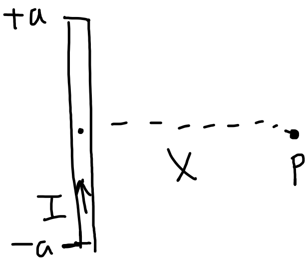
$B_{\text{of wire}} = nAl \frac{\mu_0}{4\pi} q \frac{\vec{v} \times \hat{r}}{r^2}$; $I = nqAv_d$

$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{l} \times \hat{r}}{r^2}}$$





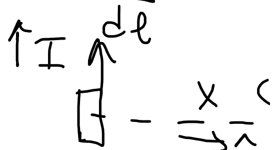
$I\, d\vec{l} \times \hat{r} \Rightarrow \vec{v} \times \hat{r}$



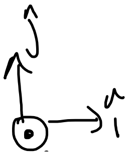
$\vec{B}_p = ?$ Length $= 2a$ I

$$\int d\vec{B} = \int \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \hat{r}}{r^2}$$

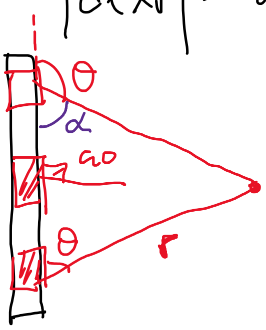
$d\vec{B} = dB(-\hat{k})$



$d\vec{l} \times \hat{r} = ?$



$|d\ell \times r| = d\ell\,(1)\sin\theta$

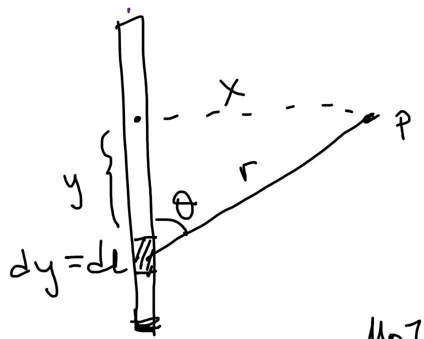


$|\hat{r}| = 1$ $\quad \int dB = \frac{\mu_0}{4\pi} I \int \frac{d\ell \sin\theta}{r^2}$

$\frac{\vec{d\ell}\times\hat{r}}{r^2} = \frac{\vec{d\ell}\times\hat{r}}{r^2}$

$\theta = \pi - \alpha$

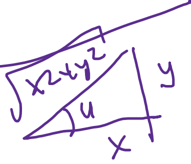
$\sin\theta = \sin(\pi - \alpha) = \sin\alpha$



$r^2 = x^2 + y^2$

$$\frac{\mu_0 I}{4\pi} \int dy \frac{\frac{x}{(x^2+y^2)^{1/2}}}{x^2+y^2}$$

$$= \frac{\mu_0 I}{4\pi} x \int_{-a}^{a} \frac{dy}{(x^2+y^2)^{3/2}} = \int \frac{x \sec^2 u\, du}{x^2 (x^2 + x^2 \tan^2 u)^{3/2}}$$



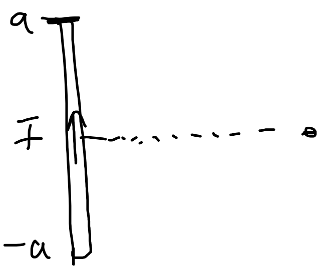
$1 + \tan^2 = \sec^2$

$\frac{y}{x} = \tan u \qquad u = \tan^{-1}\left(\frac{y}{x}\right) ; \Rightarrow \frac{dy}{x} = \sec^2 u\, du$

$$\int \frac{\sec^2 u\, du}{\sec^3 u} = \int \frac{du}{\sec u} = \int \cos u\, du \Rightarrow \underline{\sin u}$$

$$= \frac{\mu_0 I}{4\pi} x \left[\frac{1}{x^2} \frac{y}{(y^2+x^2)^{1/2}} \right]_{-a}^{a} \Rightarrow$$

$$B = \frac{\mu_0 I x}{4\pi} \left[\frac{\overbrace{a - (-a)}^{2a}}{x^2 \sqrt{x^2+a^2}} \right] = \underline{\frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2+a^2}}}$$



$B\uparrow \quad (2a)\uparrow ; \; I\uparrow ; \;$ as $x \to 0 \quad B \to \infty$?!

$x \downarrow$

If $x = 1\,m$, $2a = 1\,m$ $\quad B_p = 1$ Tesla, $I = ?$

$L = 2a$

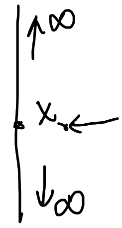
$$1 = \frac{\cancel{4\pi}\times10^{-7}}{\cancel{4\pi}} I \;\frac{1}{1\sqrt{1+0.}}$$

$I \sim 10^7 A$

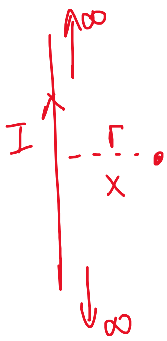
$$\frac{\mu_0 I}{4\pi}\frac{L}{x\sqrt{x^2+\left(\frac{L}{2}\right)^2}}$$

$x >> L$ $\quad \frac{L}{x} \simeq 0$ $\quad \sim \frac{\mu_0 I}{4\pi}\frac{L}{x^2}$; $\quad x << L$ $\quad \frac{x}{L} \simeq 0$

$$\sqrt{x^2 + \frac{L^2}{4}} = \sqrt{\frac{L^2}{4}\Big(\underbrace{\frac{x^2}{L^2/4}}_{\simeq 0}+1\Big)} = \frac{L}{2}$$

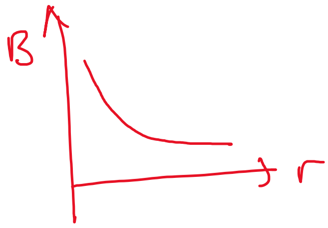


$$B = \frac{\mu_0 I}{2\pi x}$$



∞ Long wire

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$

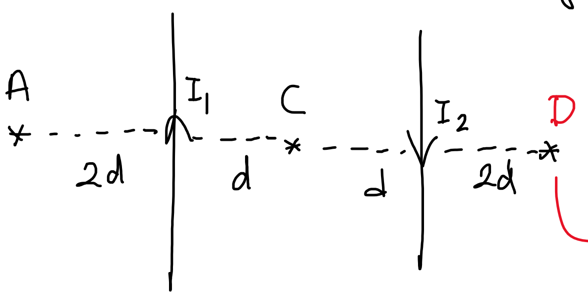


$$1T = \frac{2\times10^{-7} I}{1\mu m} \Rightarrow I = \frac{1}{2}\times 10 = 5A$$

$r = 10^{-6} m = 1\,\mu m$

ex) ∞ long 2 parallel wires, I

$B_C = ?$ $\quad B_A = ?$ $\quad B_D = ?$



$\odot\ \hat{k}$

$\otimes\ -\hat{k}$

$$B_{1A} = \frac{\mu_0 I_1}{2\pi(2d)} \odot +\hat{k} \qquad B_{1C} = \frac{\mu_0 I_1}{2\pi d} \otimes -\hat{k} \qquad B_{1D} = \frac{\mu_0 I_1}{2\pi(4d)} \otimes$$

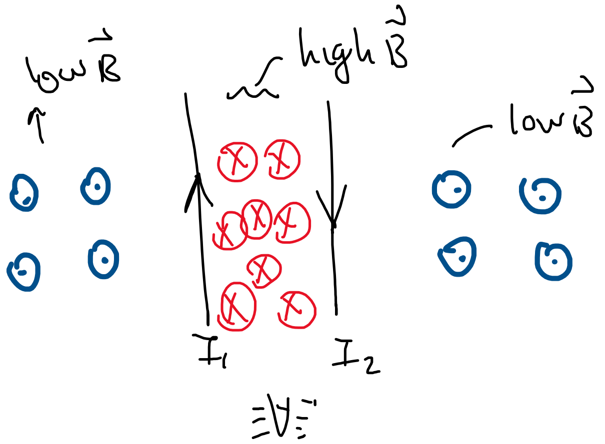
$$B_{2A} = \frac{\mu_0 I_2}{2\pi(4d)} \otimes -\hat{k} \qquad B_{2C} = \frac{\mu_0 I_2}{2\pi d} \otimes -\hat{k} \qquad B_{2D} = \frac{\mu_0 I_2}{2\pi(2d)} \odot$$

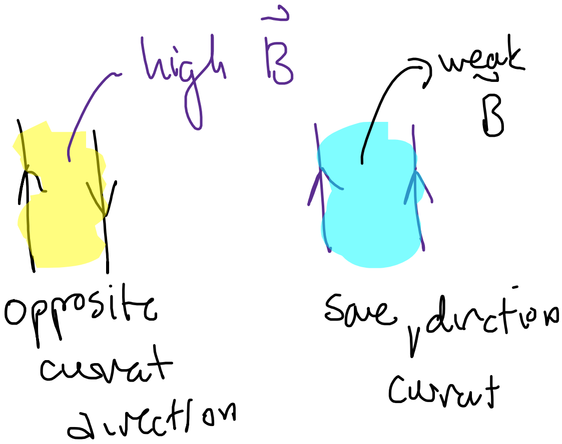
$+$ $\qquad +$ $\qquad +$

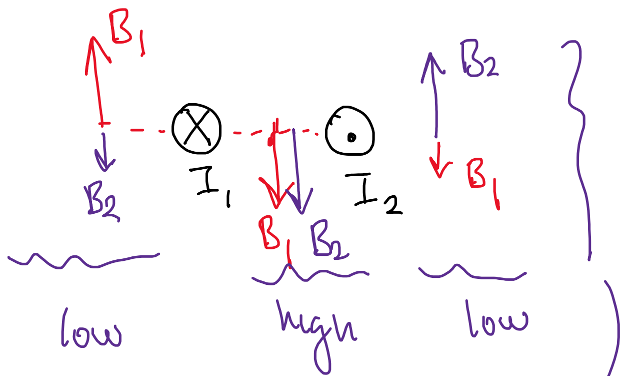
$$\frac{\mu_0}{8\pi d}(2I_1 - I_2)\hat{k} \quad \Big\{ \quad \frac{\mu_0}{2\pi d}(I_1+I_2)(-\hat{k}) \quad \Big\{ \quad \frac{\mu_0}{8\pi d}(I_1 + 2I_2)\hat{k}$$

$I_1 = I_2$

$$\frac{\mu_0 I}{8\pi d}\hat{k} \quad \Big\{ \quad \frac{\mu_0 I}{\pi d}(-\hat{k}) \quad \Big\{ \quad \frac{\mu_0 I}{8\pi d}\hat{k}$$

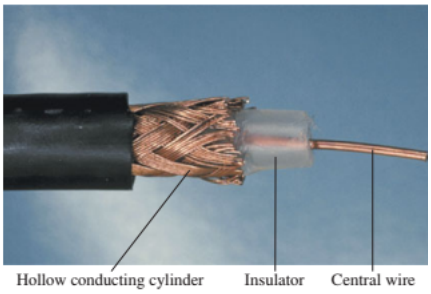


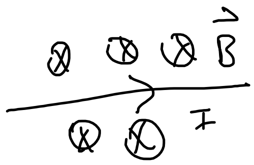




28.8 Computer cables, or cables for audio-video equipment, create little or no magnetic field. This is because within each cable, closely spaced wires carry current in both directions alon the length of the cable. The magnetic fields from these opposin currents cancel each other.

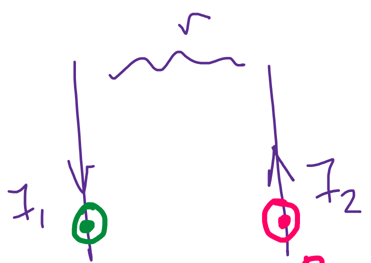
a





$$\vec{F} = I\vec{L} \times \vec{B}$$

∞ ← L

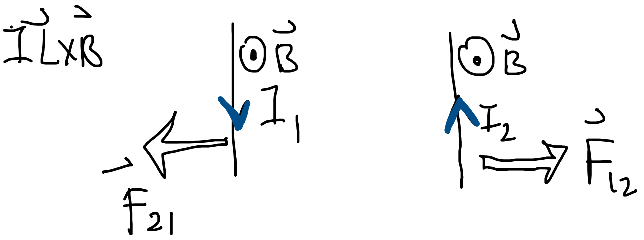


$$B_2 = \frac{\mu_0 I_2}{2\pi r} \qquad B_1 = \frac{\mu_0 I_1}{2\pi r}$$

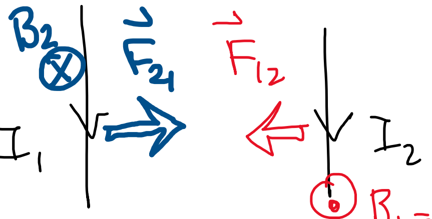
$$\vec{F}_{12} = -\vec{F}_{21}$$

$$F_{21} = I_1 L \frac{\mu_0 I_2}{2\pi r} \quad ; \quad F_{12} = I_2 L \frac{\mu_0 I_1}{2\pi r}$$

$\vec{I}\vec{L} \times \vec{B}$



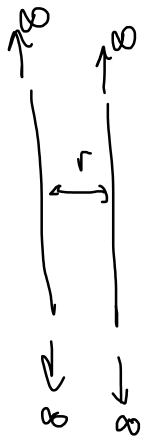
opposite I ⇒ repel
push



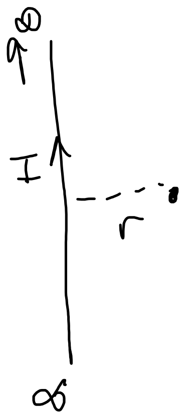
$B_1 = \frac{\mu_0 I_1}{2\pi r}$

$$\vec{F}_{12} = -\vec{F}_{21}$$

same I ⇒ attract
pull
each
other.



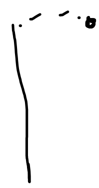
$$F = L \frac{\mu_0 I_1 I_2}{2\pi r} \Rightarrow \frac{F}{L} = \mu_0 \frac{I_1 I_2}{2\pi r}$$



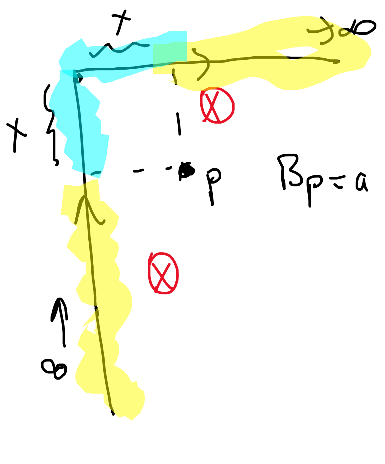
$B = \frac{\mu_0 I}{2\pi r}$

semi-infinite

$$B = \frac{1}{2} \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{4\pi r}$$

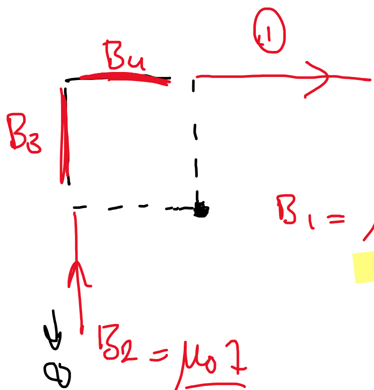


ex)



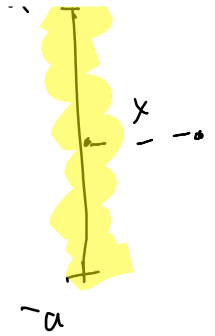
$B_{p=a}$

⇒



$B_1 = \frac{\mu_0 I}{4\pi r}$

$B_2 = \frac{\mu_0 I}{4\pi r}$



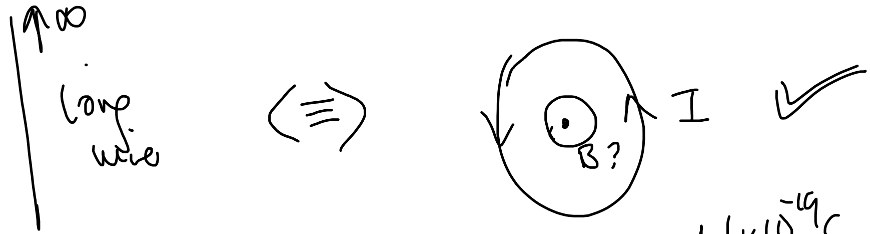
$$\frac{\mu_0 I}{4\pi}\,\frac{2a}{x\sqrt{x^2+a^2}}$$

B_3 

$$\left(\frac{1}{2}\,\frac{\mu_0 I}{4\pi}\,\frac{2a}{x\sqrt{x^2+a^2}}\right)$$

$$B_1+B_2+B_3+B_4 \Rightarrow \underbrace{\left(\frac{\mu_0 I}{4\pi r}\right)2}_{B_1+B_2} + \underbrace{2\left(\frac{1}{2}\,\frac{\mu_0 I\,2a}{4\pi\,x\sqrt{x^2+a^2}}\right)}_{B_3+B_4}$$

$\longrightarrow B_4$

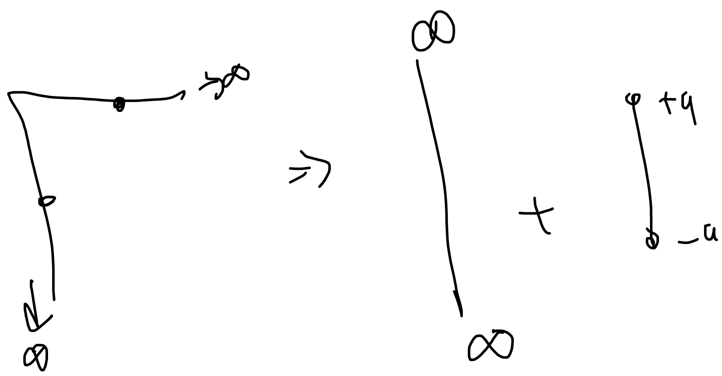


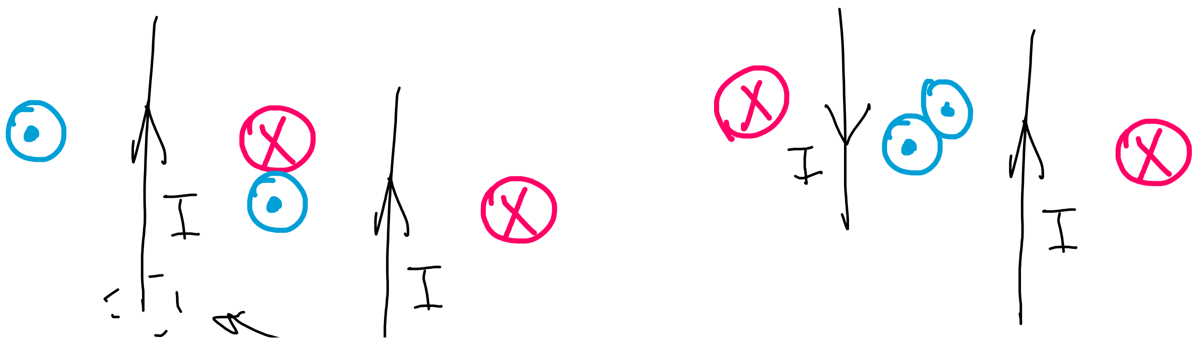
$$\text{akım} = \frac{q}{t}$$

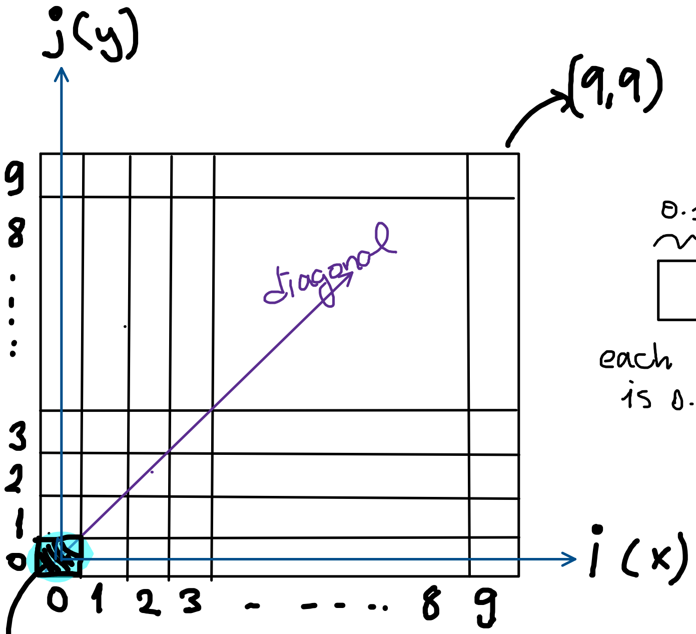
$(\#)$ sayı, $q \to 1{,}6\times10^{-19}C$

$$\hookrightarrow \frac{nAl_i q}{t} = nAv_d q$$

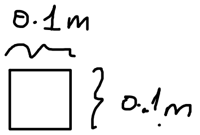
hız





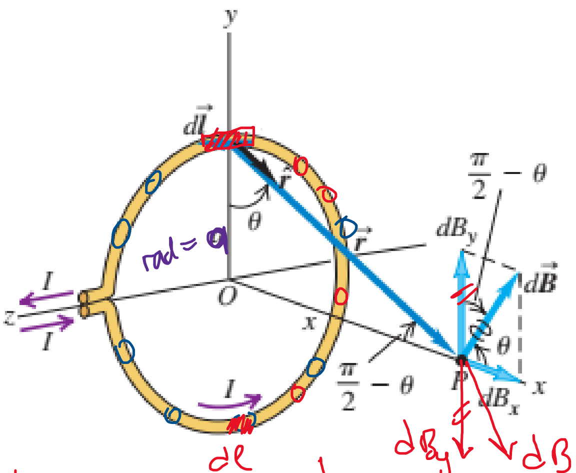


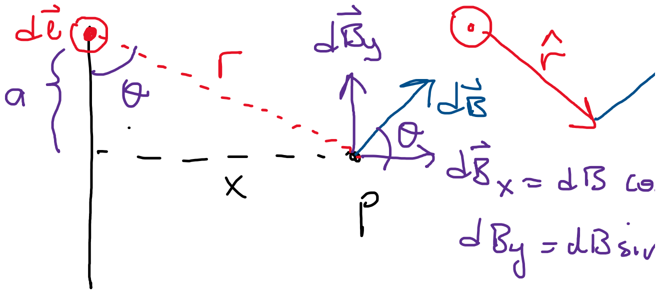
Q is at 0,0 point



each box is 0.1m in length.

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{\ell}\times\hat{r}}{r^2}$$



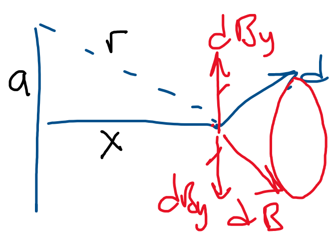


$$|d\vec{\ell}\times\hat{r}| \Rightarrow |d\vec{\ell}|\cdot|\hat{r}||\sin 90| =$$

$$\boxed{\int dB_x = B_x}$$

$$\int dB_y = B_y = 0$$

$$\int dB_y = 0$$



$$dB_x = dB\cos\theta$$

$$\int dB_x = \int \frac{\mu_0 I}{4\pi}\frac{d\ell}{r^2}\cos\theta -$$

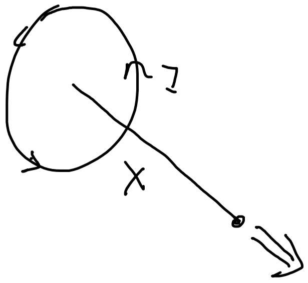
$$\int \frac{\mu_0 I}{4\pi}\frac{d\ell}{r^2}\frac{a}{r}$$

$$B_x = \frac{\mu_0 I a}{4\pi r^3}\int d\ell$$

$d\ell$; r change?

$$2\pi a$$

$$= \frac{\mu_0 I a \cancel{2}\pi a}{\cancel{4}_2 \pi r^3} = \frac{\mu_0 I a^2}{2 r^3} = \frac{\mu_0 I a^2}{2\left[a^2+x^2\right]^{3/2}}$$

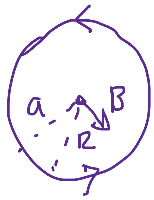


$$B_x = \frac{\mu_0 I a^2}{2(a^2+x^2)^{3/2}}$$

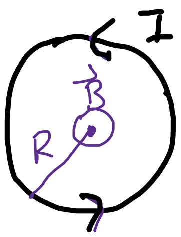
$x = 0$

$$\frac{\mu_0 I a^2}{2\left[a^2+0^2\right]^{3/2}} = \frac{\mu_0 I a^2}{2a^3}$$

$$B = \frac{\mu_0 I}{2a}$$



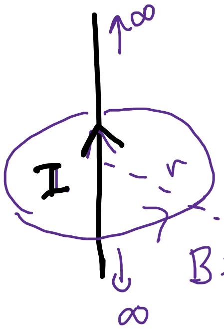
* * * $\boxed{B = \frac{\mu_0 I}{2R}}$ @ the center



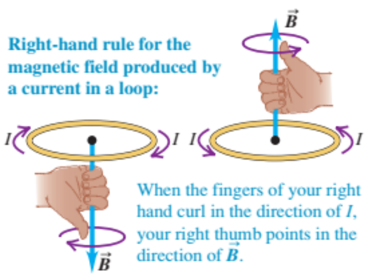
$$B = \frac{\mu_0 I}{2R}$$

circular wire

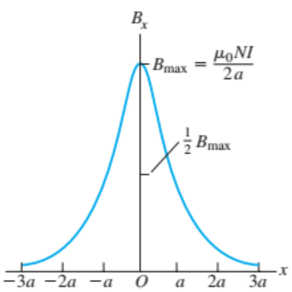
∞ Long wire

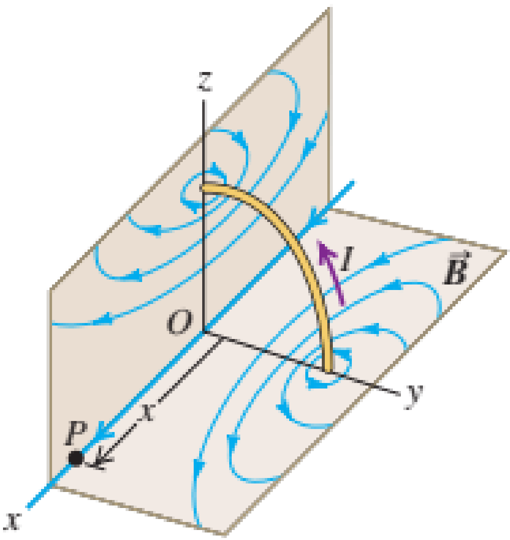


$$B = \frac{\mu_0 I}{2\pi r}$$



28.14 Graph of the magnetic field along the axis of a circular coil with N turns. When x is much larger than a, the field magnitude decreases approximately as $1/x^3$.





$$\frac{\mu_0 I}{2a} N = B$$

Maxwell's eqn.

(1) Gauss Law $\oint \vec{E}\cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$

(2) gauss Law for $\vec{B}$ $\oint \vec{B}\cdot d\vec{A} = 0$

(3) Ampere's Law: $\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$

$\oint_a^a$ closed loop

$d\vec{l}$; length

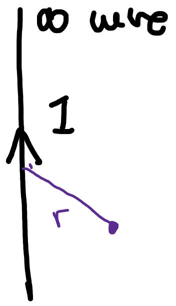
I_{enc} = amount of current enclosed by $\oint d\vec{l}$ loop.

$d\vec{B} = \mu_0 I \frac{d\vec{l}\times\hat{r}}{r^2}$

$B = \mu_0 I \frac{1}{\text{length}}$

$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$

$\Rightarrow$ B length $= \mu_0 I$

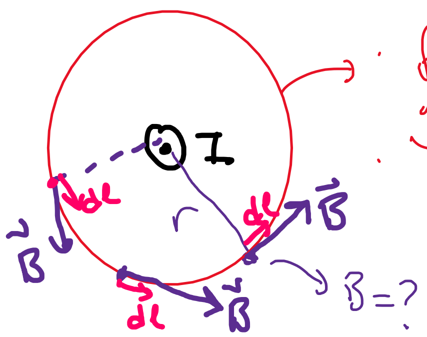


previously

$B = \frac{\mu_0 I}{2\pi r}$ ✓

re-calculate $\vec{B}$ with Ampere's Law

TOP VIEW

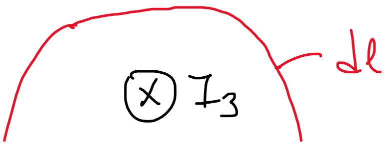


$\vec{B}\cdot d\vec{l} = B\,dl\cos\theta$ ($\theta \to 0°$)
$= B\,dl$

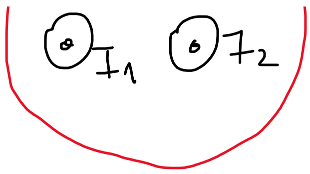
$\oint B\,dl = B\oint dl = B\,2\pi r$

B const? YES why? Because r distance away from cw

$\oint \vec{B}\cdot d\vec{l} = B\,2\pi r = \mu_0 I_{enc} = \mu_0 I$ $\quad B = \frac{\mu_0 I}{2\pi r}$ ✓



$\oint \vec{B}\cdot d\vec{l} = \mu_0 (I_1 + I_2 - I_3)$

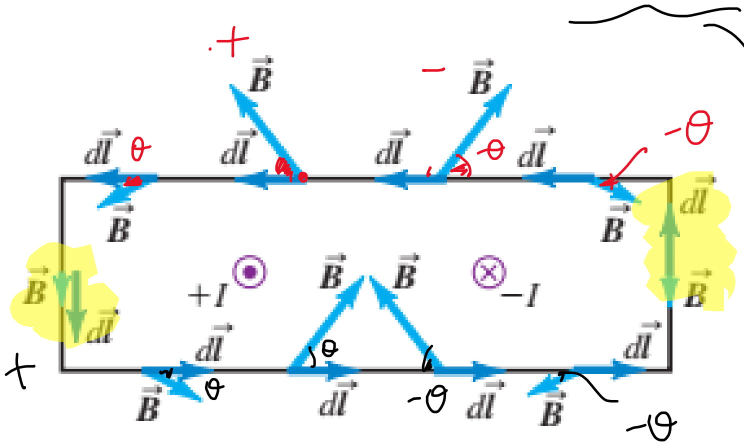


$B \neq$ const

if $I_1 + I_2 - I_3 = 0$

$B = 0 \qquad \oint \vec{B} \cdot d\vec{l} = 0$

Ampere's Law $\Rightarrow$ $\underbrace{\oint \vec{B} \cdot d\vec{l}}$ $= \mu_0 I_{enc} = \mu_0 (I - I) = \underline{\underline{0}}$



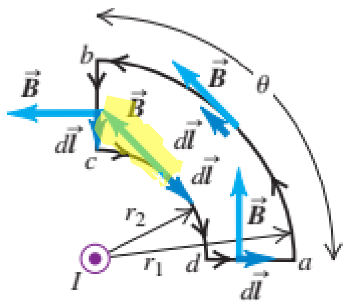
$+1{,}2 \; -1{,}2 \; +9 \; -9$

$+ \cdots = \underline{\underline{0}}$

$\vec{B} \cdot d\vec{l} = B\,dl \cos\theta$

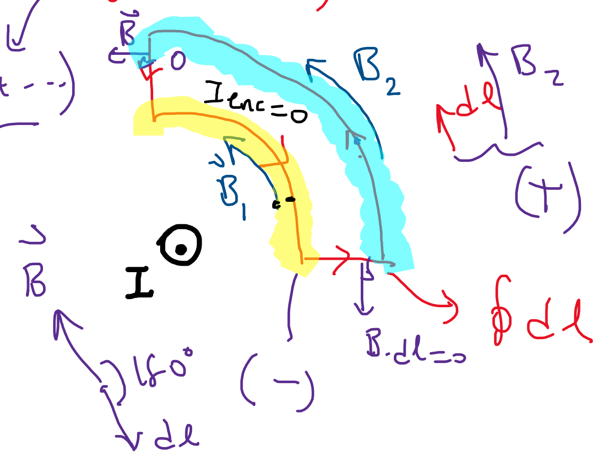
(c) An integration path that does not enclose the conductor

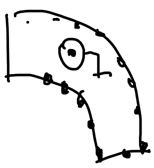
Result: $\oint \vec{B} \cdot d\vec{l} = 0$



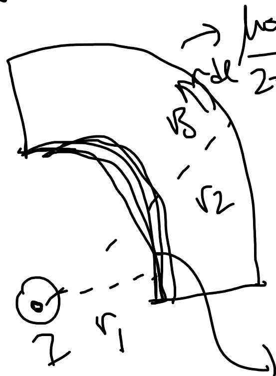
$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} = 0$

$\underbrace{(+1{,}2 - 1{,}2 + \cdots)}_{0}$

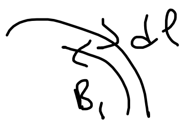


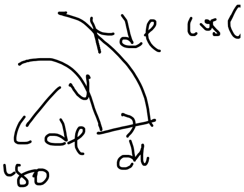


$\oint \vec{B} \cdot d\vec{l} \neq 0$



$\underbrace{\frac{\mu_0 I}{2\pi r_2} \; \frac{2\pi r_2}{4} (+1)}_{+}$





$\underbrace{\cos 180}$

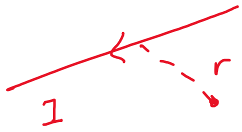
cos180

$\underbrace{\frac{\mu_0 I}{2\pi r_1} \; \frac{2\pi r_1}{4} (-1)}_{-}$

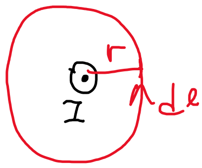
$B_1 = \frac{\mu_0 I}{2\pi r_1} \qquad B_2 = \frac{\mu_0 I}{2\pi r_2}$

$\vec{B}$ of ∞ Long wire Inside / outside?

Long wire is a cylinder with radius R



$r > R$ (outside)



$$\int \vec{B} \cdot d\vec{l} = \mu_0 I_{en}$$

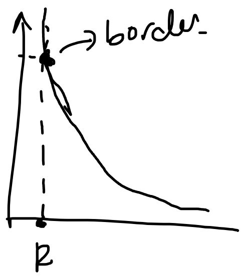
$$\int B\, dl \cos 0$$

$$B \underbrace{\int dl}_{} = \mu_0 I$$

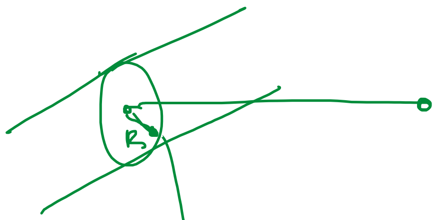
$$B\, 2\pi r = \mu_0 I$$

$$B(r > R) = \frac{\mu_0 I}{2\pi r} \text{ outside}$$

$$B = \frac{\mu_0 I}{2\pi r}$$



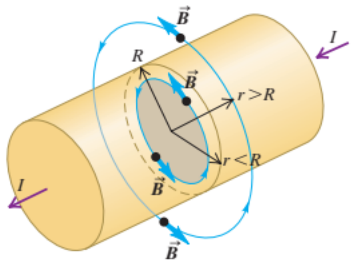
$$B(r = R) = \frac{\mu_0 I}{2\pi R}$$



$$\frac{\mu_0 I}{2\pi r}$$

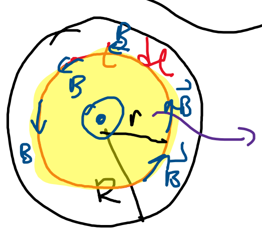
$$\frac{\mu_0 I}{2\pi R}$$

28.20 To find the magnetic field at radius $r < R$, we apply Ampere's law to the circle enclosing the gray area. The current through the red area is $(r^2/R^2)I$. To find the magnetic field at radius $r > R$, we apply Ampere's law to the circle enclosing the entire conductor.



$r < R$ (inside the wire)

$$\mu_0 I_{enc} = \oint \vec{B} \cdot d\vec{l}$$



I_0 = total current

I_{enc} = ?

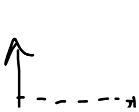
$$\frac{\pi r^2}{\pi R^2} = \frac{I_{enc}}{I_0} = \frac{\text{Area en}}{\text{Total ar}}$$

$$\boxed{I_{enc} = I_0 \frac{r^2}{R^2}}$$

$$\mu_0 I_0 \frac{r^2}{R^2} = \oint \vec{B} \cdot d\vec{l}$$

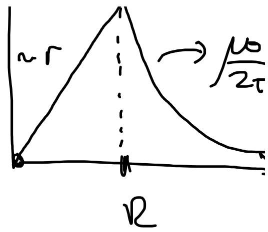
$$\oint B\, dl \cos 0 \; (=1)$$

$$\mu_0 I \frac{r^2}{R^2} = B \underbrace{\oint dl}_{2\pi r}$$

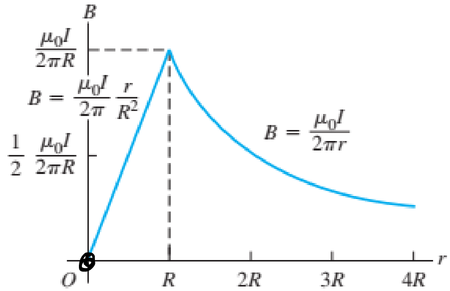


$$\frac{\mu_0 I r^{\cancel{2}}}{R^2} = B\, 2\pi \cancel{r} \Rightarrow \boxed{B(r<R) = \frac{\mu_0 I}{2\pi R^2} r}$$

inside

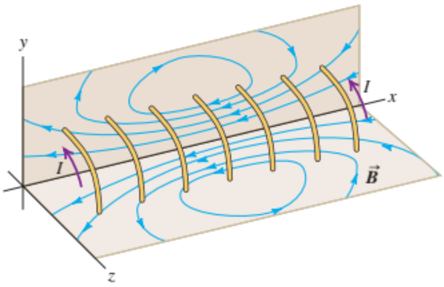


28.21 Magnitude of the magnetic field inside and outside a long, straight cylindrical conductor with radius R carrying a current I.

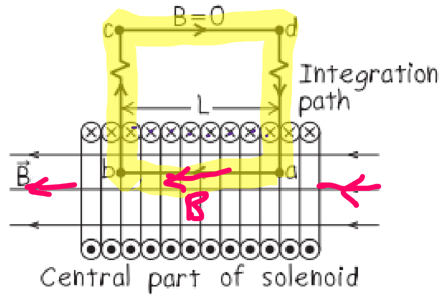


$\vec{B}$ of SOLENOID

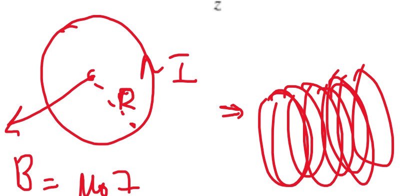
28.22 Magnetic field lines produced by the current in a solenoid. For clarity, only a few turns are shown.



Our sketch for this problem.

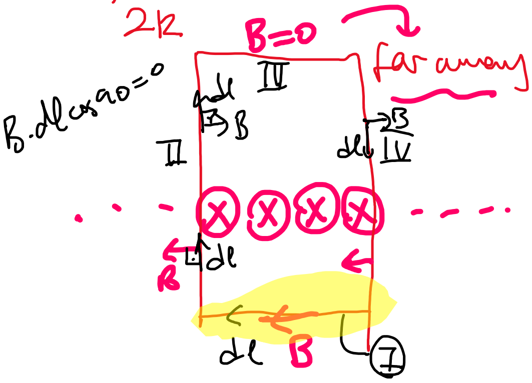


$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 I N$$

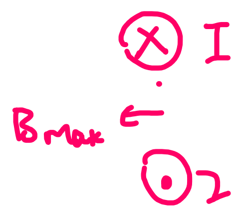


$B = \frac{\mu_0 I}{2R}$

$B = N \frac{\mu_0 I}{2R}$?!



$\longleftrightarrow B \simeq 0$



$$\oint \vec{B}\cdot d\vec{l}$$

$$\int_I \vec{B}\cdot d\vec{l} + \int_{II} B\cdot dl + \int_{III} \ldots$$

$$\int_{IV} (B=0)\cdot dl$$

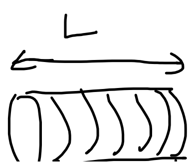
$$\int_I \vec{B}\cdot d\vec{l} = BL + 0 + 0 + 0$$

$$\int B\, dl \quad B\int dl = BL$$

$\cos 90 =$

$$BL = \mu_0 I N$$

$$B = \frac{\mu_0 I N}{L}$$



N = no of

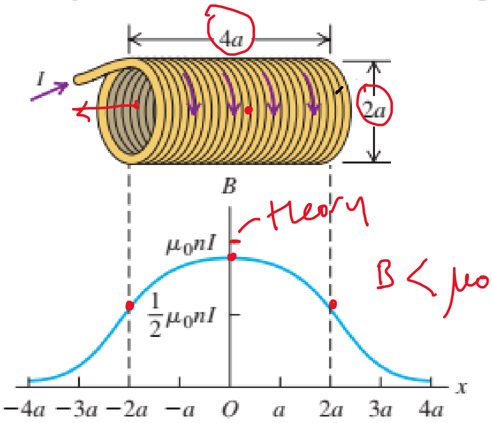
L

turns

$B\uparrow;\ I\uparrow,\ N\uparrow,\ L\downarrow$ $\quad \boxed{B = \mu_0 I \frac{N}{L}}$ solenoid ideal

$\frac{N}{L} = \underline{n}$ no. of turns per length

28.24 Magnitude of the magnetic field at points along the axis of a solenoid with length $4a$, equal to four times its radius a. The field magnitude at each end is about half its value at the center. (Compare with Fig. 28.14 for the field of N circular loops.)

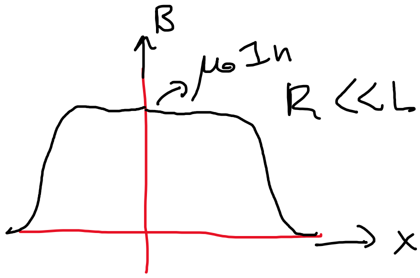


$\mu_0 I \frac{N}{4a}$

$B < \mu_0 n I$

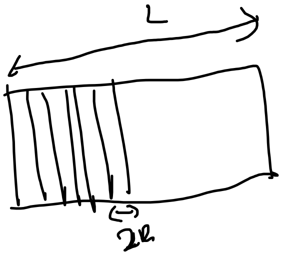
$B = \mu_0 I n$

$\Rightarrow$



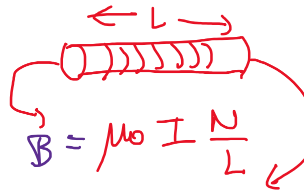
$R \ll L$

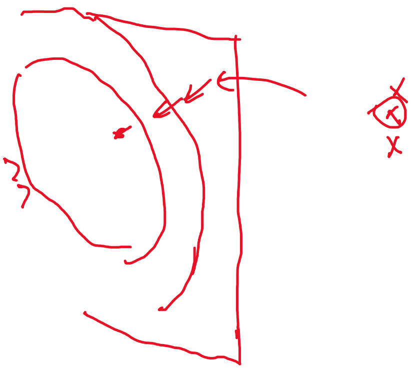
$L \gg R$

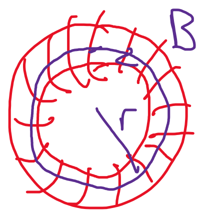


$\frac{L}{2R} =$ no. of turns

$\vec{B}$ field of TOROIDAL SOLENOID.



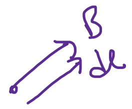




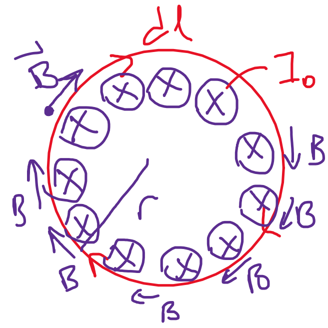
$B = \mu_0 I \frac{\#\text{ of}}{\text{Le}}$

$B = \mu_0 I \frac{N}{2\pi}$

Ampere's Law: $\oint \vec{B}\cdot d\vec{l} = \mu_0 \underbrace{I_{enc}}_{N I_0}$ $\Rightarrow$ $\underbrace{\oint B\,dl}_{\text{const.}} = \mu_0 N I$



$\vec{B}\cdot d\vec{l} = \underset{\downarrow\ \text{const}}{B}\,dl\,\cos 0 = 1$

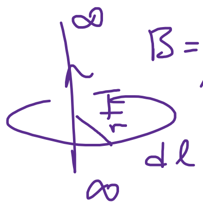


$B \underbrace{\oint dl}_{2\pi r} = \mu_0 N I_0$

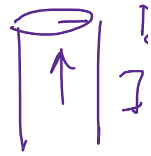
$$\boxed{B = \mu_0 \frac{I_0 N}{2\pi r}}$$

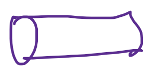
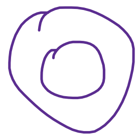
$B\uparrow;\ I_0\uparrow,\ \underbrace{N\uparrow\ \ r\downarrow}_{\text{thin wires}}$

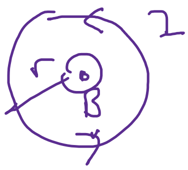
$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$ $\rightarrow$



$B = \mu_0 \frac{I}{2\pi r}$ ✓

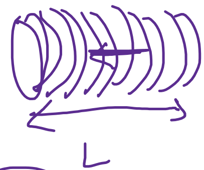


 ✓ ;  ✓



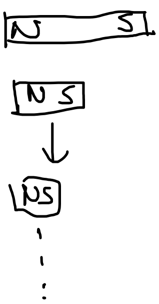
$B = \mu_0 \frac{I}{2r}$

single loop



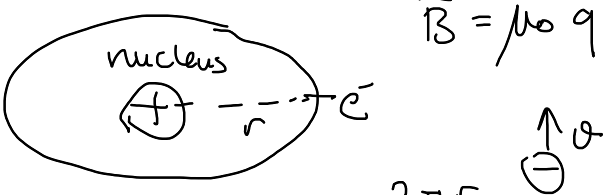
$B = \mu_0 I \frac{N}{L}$ $\neq$

BOHR MAGNETON



atom has magnetic property

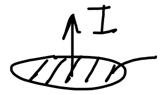
$\vec{B} = \mu_0 q \frac{\vec{v}\times\hat{r}}{r^2}$



UCM $\Rightarrow$ $v = \frac{2\pi r}{T}$ $\rightarrow$ $T = \frac{2\pi}{\ }$

$\mu = IA$

magnetic moment



area =

$\rightarrow$ $\frac{Q}{\text{time}} = \frac{q_e}{T} = \frac{q}{2}$

$\mu = IA = \frac{q_e}{2\pi r} v\, \pi r^2 = \frac{q_e v r}{2} \frac{m}{m}$ phys I

$L = r \times p$

$\mu = \frac{q_e L}{2m}$ angular mom.

$L =$ const for e^- quantum $= \frac{h}{2\pi}$ → planck const.

$L = r m v$

$\mu = \frac{q_e}{2m} (r m v) L = \frac{q_e h}{4\pi m} \equiv$ const. for $e^- = 9.3 \times 10^{-}$

↳ amount of $\vec{B}$ that e^- creates!

Atoms

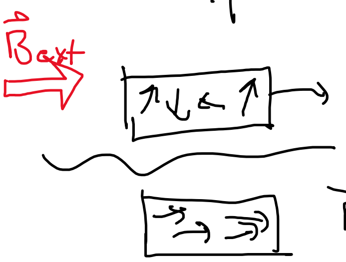
ferromagnetic

(Iron like)

(Iron, Nickel, Co...)

→ make magnets.

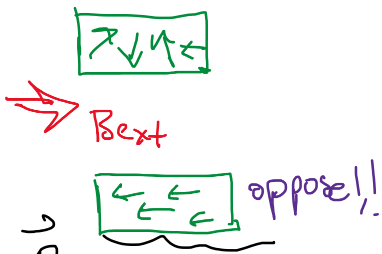
keep magnetism when exposed to $\vec{B}$ external.



↳ keep it magnets.

diamagnetic

$\vec{\mu}$ opposes $\vec{B}_{ext}$; they are not permanent

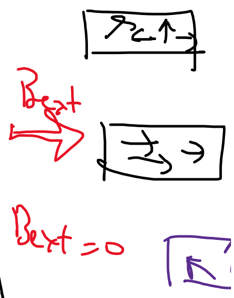


$\vec{B}_{ext} + \sum B_{opposition} \cong 0$

paramagnetic

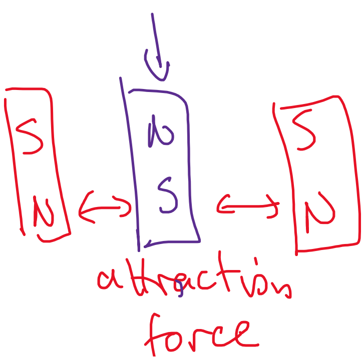
$\vec{\mu}$ moments align with but goes

$B_{ext} = 0$

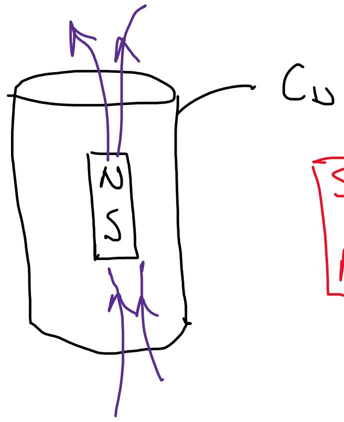


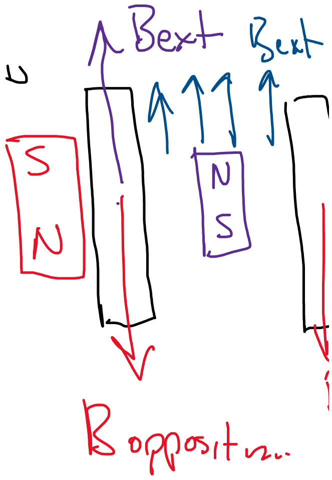
Diamagnetic

Copper



N





magnet moves
slowly inside a cu tube.

Electromagnetic Induction (induksiyon / etkileme)

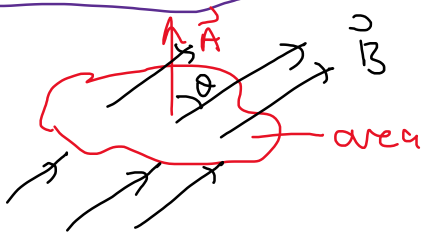
2nd Maxwell equ $\underbrace{\oint \vec{B} \cdot d\vec{A}}_{\text{magnetic flux}} = 0$

$\Phi_B = \underbrace{\int \vec{B} \cdot d\vec{A}}_{\text{magnetic flux (akı)}}$

$$\boxed{-\frac{d\Phi_B}{dt} = \overset{(EMF)}{\mathcal{E}}}$$

* change in magnetic flux
* produces potential d

FLUX was:



$\int \vec{B} \cdot d\vec{A} = \underset{\text{const.}}{\vec{B}} \cdot \vec{A}$

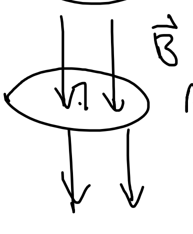
$\vec{B}, d\vec{A}$ vectors.

$B\,dA \cos\theta$ → magnetic field, area, angle

$$\left[\frac{T\,m^2}{s} = \text{Volt}\right]$$

$$\left[T = \frac{Vs}{m^2}\right]$$

$-\frac{d\phi}{dt}$ → opposing



$\frac{dB}{dt} = 0.02\ T/s$ increasing

$A = 0.012\ m^2$

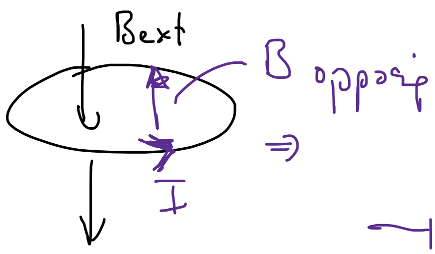
EMF direc & mag

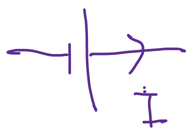
$\mathcal{E} = -\frac{d\Phi_B}{dt}$

$\frac{d\Phi_B}{...} = dB\,A$

$$\frac{d\Phi_B}{dt} = \frac{dB}{dt}A + B\cancel{\frac{dA}{dt}}$$

$\cancel{\frac{dA}{dt}} \to 0$ Area

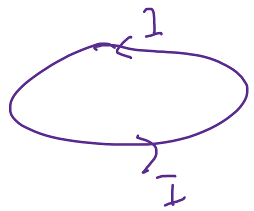




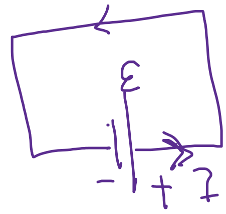
$$= 0.02\,\frac{T}{s}\ 0.012\ m^2$$

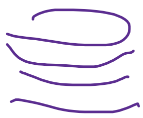
$$\varepsilon = 24\times10^{-5}\ V$$

$$\varepsilon = 0.24\ MV$$



$\equiv$

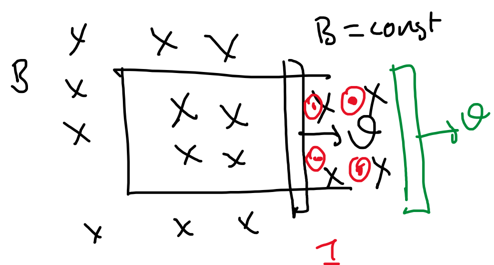




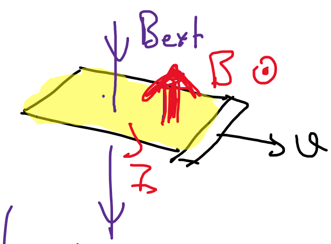
$N\uparrow$

$$-N\frac{d\Phi}{dt} = \varepsilon$$

SLIDE WIRE GENERATOR



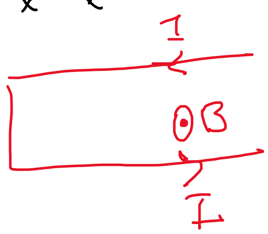
$$-\frac{d\tilde{\Phi}}{dt} = \varepsilon$$



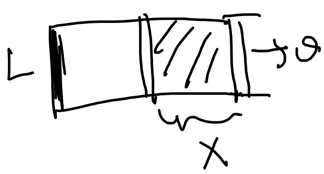
$\Phi = BA$

Const ↑

$\int B \cdot dA$



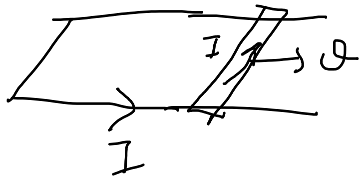
$$\frac{d\Phi_B}{dt} = \frac{d}{dt}(BA) = \cancel{\frac{dB}{dt}}A + B\underbrace{\frac{dA}{dt}}_{vL} = \varepsilon$$

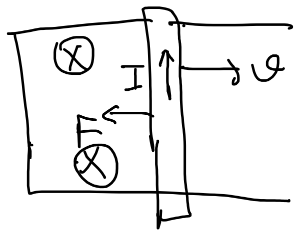


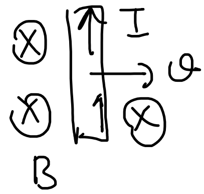
area $= A = xL$

$$\frac{dA}{dt} = \underbrace{\frac{dx}{dt}}_{v}L + x\cancel{\frac{dL}{dt}}^{\,0}$$

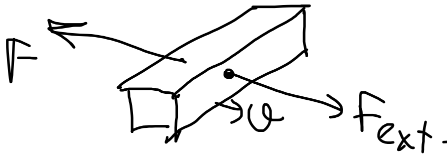
$$\boxed{\varepsilon = BvL}$$

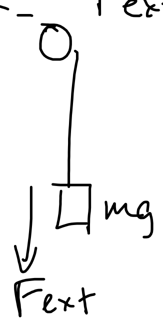






$\vec{F} = I\vec{L} \times \vec{B}$





$F_{ext} = ILB$

$v = \text{const}$

phys I $F_{ext} \cdot v \equiv$ power

$\downarrow$

$ILBv$ = power

Power = $\frac{\text{Energy}}{\text{time}}$ =

$= \frac{F\,x}{t} =$

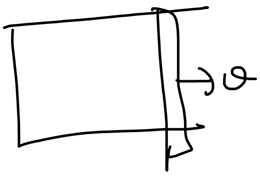
phys II

Power = $IV = I^2R$

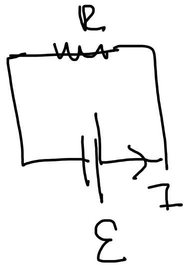
$\downarrow$

ε

$$\boxed{I\varepsilon = \frac{\varepsilon^2}{R} = ILBv}$$



$\Rightarrow$



$\varepsilon = BvL$

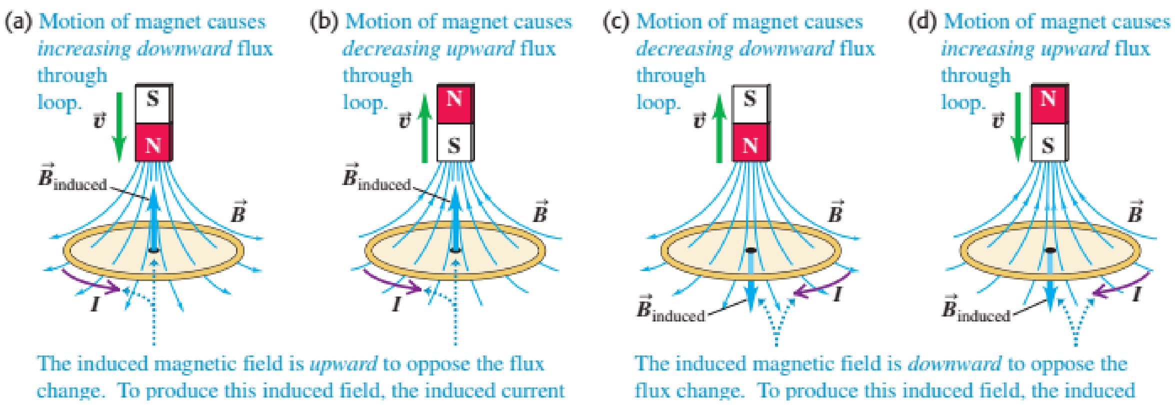
$v = 10$ m/s $\quad L = 1$ m $\quad B = 0.1$ T $\quad R = 10\Omega$

$$\frac{\varepsilon^2}{R} = \frac{B^2v^2L^2}{R} = \text{power}$$

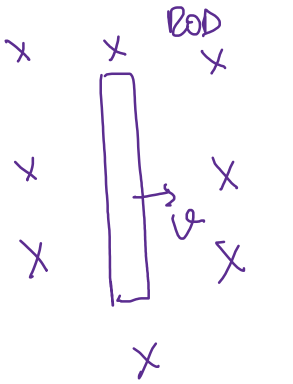
$$\frac{(BvL)^2}{R} = \text{power} = \frac{(0.1 \times 10 \times 1)^2}{10\Omega} = \underline{\underline{0.1 \text{ watt}}}$$

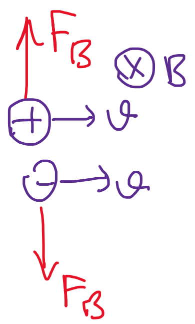
$$\frac{d}{dt}\Phi_B = -\varepsilon = \frac{d}{dt}(\vec{B}\cdot\vec{A})$$

29.14 Directions of induced currents as a bar magnet moves along the axis of a conducting loop. If the bar magnet is station[…] is no induced current.

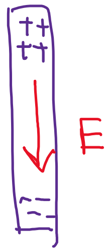


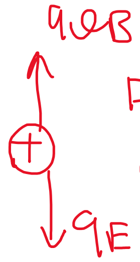
must be *counterclockwise* as seen from above the loop. current must be *clockwise* as seen from above the loop.





$\vec{F}_B = q\vec{v}\times\vec{B}$



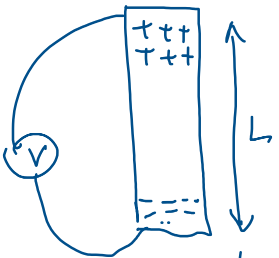


$F_B = F_E$

$qvB = qE$

$\boxed{E = vB}$

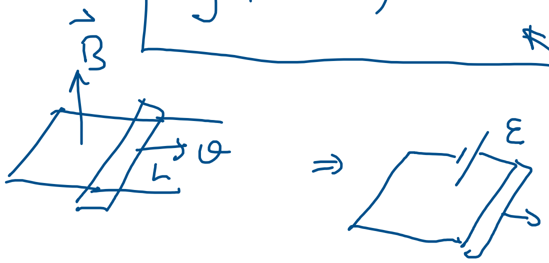
$\vec{E} = \vec{v}\times\vec{B}$



$$EL = \Delta V = \mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}$$

$\vec{v}\times\vec{B}$

$$\boxed{\oint(\vec{v}\times\vec{B})\cdot d\vec{l} = \mathcal{E}} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}$$

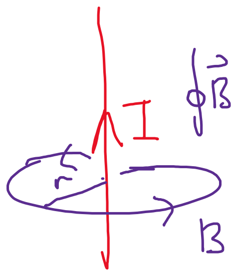


$\mathcal{E} = BvL$

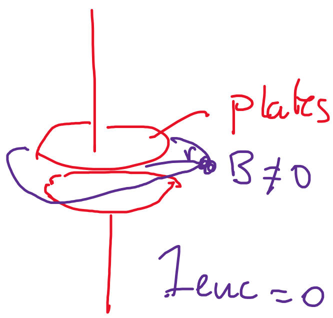
GENERALIZED AMPERE'S LAW

Ampere's Law

$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$

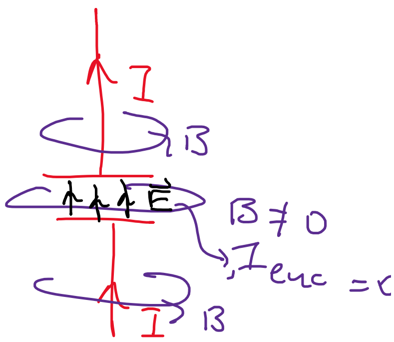


$B = \dfrac{\mu_0 I}{2\pi r}$



$I_{enc} = 0$

between plates.



$I_{enc} = 0$

$$\boxed{\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \varepsilon_0 \frac{d}{dt}(\vec{E}\cdot\vec{A})}$$

3rd Maxwell Law

$\vec{E}\cdot\vec{A} = \Phi_E$ = Electric

ε_0 = electric permittivity of sp

8.85×10^{-12} (geçirgenlik)

(...)

$\mu_0 \varepsilon_0 = \frac{1}{c^2}$; c = speed of light

μ_0 = magnetic " "

$\mu_0 = 4\pi \times 10^{-7}$ (...)

MAXWELL EQUATIONS for ElectroMagnetism

① $\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$

② $\oint \vec{B} \cdot d\vec{A} = 0$

③ $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_{enc} + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$; $\Phi_E = \vec{E} \cdot \vec{A}$

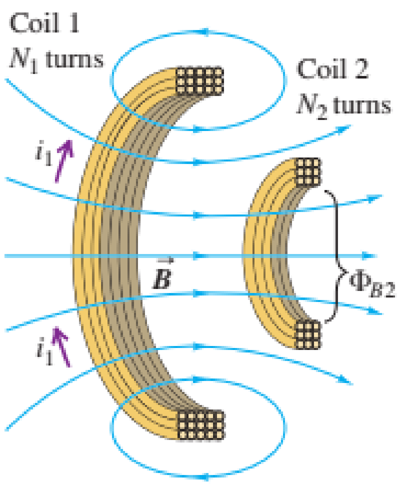
④ $\underbrace{\oint \vec{E} \cdot d\vec{l}}_{\mathcal{E}} = -\frac{d}{dt} \Phi_B$; $\Phi_B = \vec{B} \cdot \vec{A}$

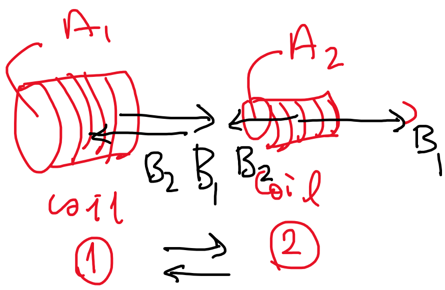
End of Ch 29

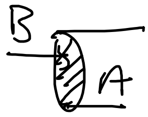
Chapter 30 ⇒ INDUCTANCE

→ karşılıklı

Mutual inductance: If the current in coil 1 is changing, the changing flux through coil 2 induces an emf in coil 2.







I_1 in coil 1 form $\vec{B}$; goes into c 2nd coil

$\mathcal{E}_2 = -N_2 \frac{d\Phi_{B2}}{dt}$

no of turns

proportionality constant

$M_{21} i_1 = N_2 \Phi_1$

$\left\{ M = \text{mutual inductance} = \frac{N\Phi}{i} \right\}$

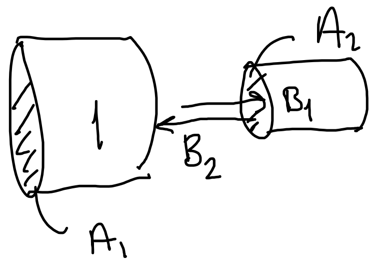
$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt}$; $\frac{d}{dt}(\Phi_{B2} =$

If coil-2 has i_2 current $\Rightarrow B \Rightarrow$ affect 1st coil.

$$\mathcal{E}_1 = -M_{12}\frac{di_2}{dt} \qquad * \left\{ M_{12} = M_{21} = \frac{N_2\Phi_2}{i_1} = \frac{N_1\Phi_1}{i_2} \right.$$

$$\left[M = \frac{Tm^2}{A} = \frac{Wb}{A} = \text{Henry} \right] \qquad \left[H = \frac{Wb}{A} = \frac{Vs}{A} = \Omega s = \right.$$

$$M = \frac{N_2\Phi_{B2}}{i_1} = \frac{N_1\Phi_{B1}}{i_2} \Rightarrow$$

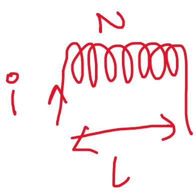


$B_1 A_2 =$

$\phi_{B1} =$

$$M = \frac{N_2 B_1 A_2}{i_1} = \frac{N_1 B_2 A_1}{i_2}$$

solenoid



$$\boxed{B = \mu_0 i \frac{N}{L}}$$

$$M = \frac{N_2\left(\mu_0 \cancel{i_1} \frac{N_1}{L_1}\right) A_2}{\cancel{i_1}} =$$

$$M = \mu_0 N_1 N_2 \frac{A_2}{L_1}$$

ex) $N_1 = 1000 \quad N_2 = 10 \quad A = 10\,cm^2 \quad l = 0.5\,m \quad \mu_0 = 4\pi \times 10^{-7}$

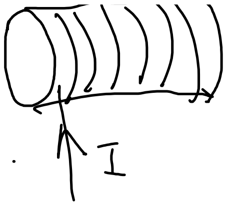
$M = 25\times10^{-6}\,H \qquad \frac{di_2}{dt} = 2\times10^6\,A/s \qquad \mathcal{E} = ?$

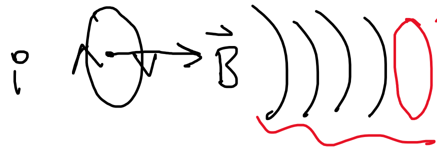
$$\mathcal{E} = -M\frac{di_2}{dt} \Rightarrow -25\times2\times10^{-6+6} = -50\,V$$

$$\mathcal{E} = -N\frac{d\Phi_B}{dt} \Rightarrow \frac{d}{dt}(Mi) = \frac{d}{dt}(N\Phi_B)$$

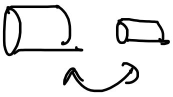
$$\boxed{\mathcal{E} = -M\frac{di}{dt}} \; *$$

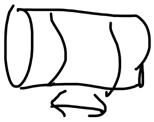
SELF INDUCTANCE





affected by 1st coil
feel $\vec{B}$ before
the current reach
the loop.

mutual inductance 

self inductance 

$$M = \frac{N\phi_B}{i}$$

$$L = \frac{N\phi_B}{i}$$

$$\mathcal{E} = N_2 \frac{d\phi_{B2}}{dt} = M \frac{di_1}{dt}$$

$$M = \mu_0 N_1 N_2 \frac{A}{L}$$

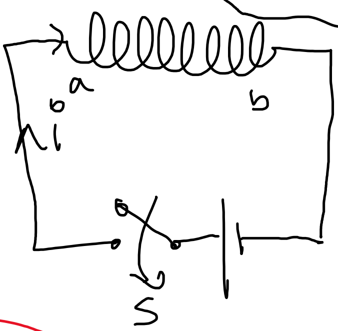
self inductance

$$Li = N\phi_B$$

$$L\frac{di}{dt} = N\frac{d\phi_B}{dt}$$

$-\mathcal{E}$

potential difference between a-b



$$\underline{\underline{V_{ab} = \mathcal{E}}}$$

$$\boxed{-L\frac{di}{dt} = \mathcal{E}} \longleftrightarrow iR = \mathcal{E}$$

R

$$[A\Omega = V]$$

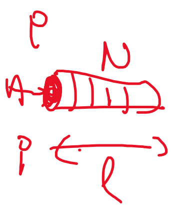
$$i = \left(\frac{V}{\Omega}\right)$$

$$L\frac{i}{t} = \mathcal{E}$$

$$H\frac{V}{\Omega s} = V$$

$$\boxed{H = \Omega s}$$

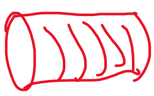
$$L = \frac{N\phi_B}{i}$$



$$B = \mu_0 i \frac{N}{\ell}$$

$$L = \frac{N\left(\mu_0 i \frac{N}{\ell}\right)A}{i}$$

$$\Leftarrow L = \mu_0 N^2 \frac{A}{\ell}$$



$$L = \mu_0 N^2 \frac{A}{\ell}$$

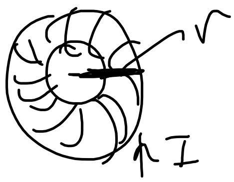
$N = 1000$

$A = 1\,cm^2$

$\ell = 1\,mm$

$$L = 4\pi \times 10^{-7} \times 10^{6} \times \frac{10^{-4}\,m^2}{10^{-3}\,m} = 4\pi \times 10^{-2} = \underline{\underline{0.12}}$$

TOROID



$$B = \mu_0 i \frac{N}{2\pi r}$$

$$L = \frac{N\phi_B}{i} = \mu_0 \frac{N^2 A}{2\pi r} \quad \text{(toroid)}$$

ex) $N = 1000$, $A = 5\ cm^2$, $r = 10\ cm$ $\Big\}$ $L = $

C — R — L

$\Delta V = \frac{Q}{C}$ $\Delta V = iR$ $\Delta V = L\frac{di}{dt}$

$\frac{\text{Energy}}{\text{time}} = \text{power}$

$[J = W s]$

$\text{Energy} = \frac{Q^2}{2C} = \frac{QV}{2}$

$\text{Power} = I^2R = IV$

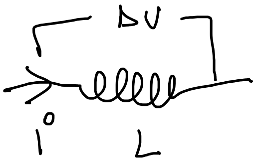
$\text{power} = p = IV = L\frac{di}{dt}i$

$\int p\, dt = L\int i\, di$

$p\,\Delta t = \frac{Li^2}{2} \equiv E$

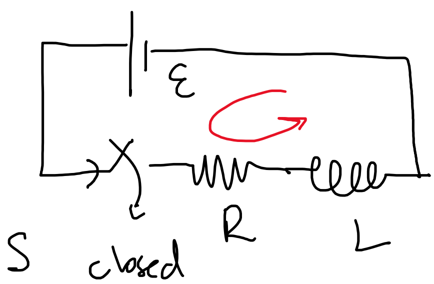
$\text{Energy} = \text{power} \cdot \text{time} = p\,\Delta t$

$\text{power} = \frac{Li^2}{2\Delta t}$



$-L\frac{di}{dt} = \Delta V$

RL circuit



Kirchoff voltage Law

$$\mathcal{E} - V_R - V_L = 0$$

$$\mathcal{E} - iR - L\frac{di}{dt} = 0$$

$R \quad di \qquad = (\mathcal{E} - i) - L\ di$

$$\frac{R}{L} dt = \frac{di}{\left(\frac{\varepsilon}{R} - i\right)} \qquad \left(\frac{\varepsilon}{R} - i\right) = \frac{\ldots}{R\, dt}$$

$$\int_{t_i}^{t_f} -\frac{R}{L} dt = \int_{i(i)}^{i_f} \frac{di}{\left(i - \frac{\varepsilon}{R}\right)} \Rightarrow -\frac{R}{L} \Delta t \Big|_{t_i=0}^{t_f=t} = \ln\left(i - \frac{\varepsilon}{R}\right)\Big|_{i(t=0)}^{i(t)}$$

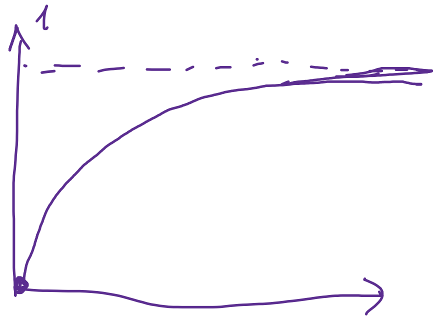
$$\boxed{-\frac{R}{L} t = \ln\left(i(t) - \frac{\varepsilon}{R}\right) - \ln\left(\cancel{i(0)}\ldots\right.}$$

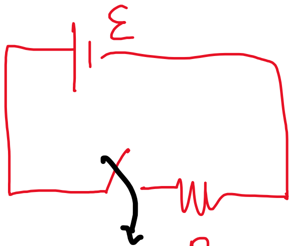
$i(t=0) = 0$

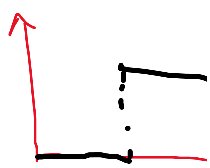
$$-\frac{R}{L} t = \ln\left[\frac{i(t) - \frac{\varepsilon}{R}}{-\frac{\varepsilon}{R}}\right]$$

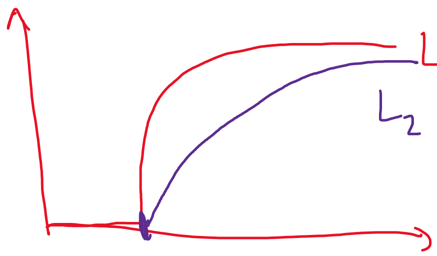
$$\left(-\frac{\varepsilon}{R}\right) e^{-\frac{R}{L}t} = i(t) - \frac{\varepsilon}{R}$$

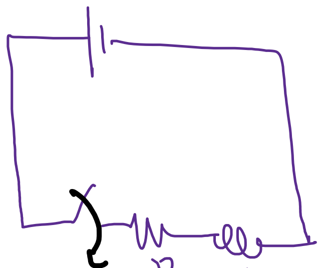
$$i(t) = \frac{\varepsilon}{R} - \frac{\varepsilon}{R} e^{-\frac{R}{L}t} = \frac{\varepsilon}{R}\left(1 - e^{-\frac{R}{L}t}\right) = i(t)$$

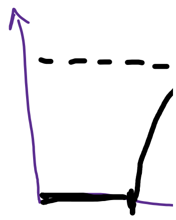












$$i(t) = \frac{\varepsilon}{R}\left(1 - e^{-\frac{R}{L}t}\right)$$

$$\varepsilon = L\frac{di}{dt} = L\frac{i}{t} = i\ldots$$

$$\varepsilon = iR = iR \qquad \frac{L}{R} = \ldots$$

$$= \frac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right) \quad ; \quad \tau = \text{time const}$$

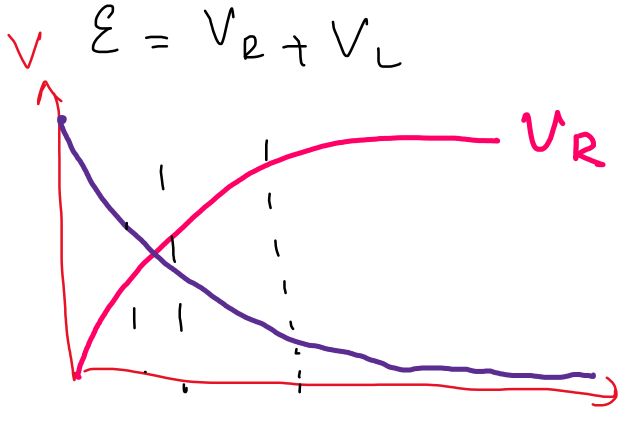
$$\frac{R}{L} =$$

$$i(t) = \frac{\varepsilon}{R}\left[1 - e^{-t/(L/R)}\right] \qquad \tau = \frac{L}{R} = [\text{sec}]$$

$$V_R = iR = \frac{\varepsilon}{\cancel{R}}\left(1 - e^{-\frac{R}{L}t}\right)\cancel{R}$$

$$V_L = L\frac{di}{dt} = L\,\frac{\varepsilon}{R}\left(0 - \left(-\frac{R}{L}\right)e^{-\frac{R}{L}t}\right) = \varepsilon\, e^{-\frac{R}{L}t}$$

$$\varepsilon = V_R + V_L$$

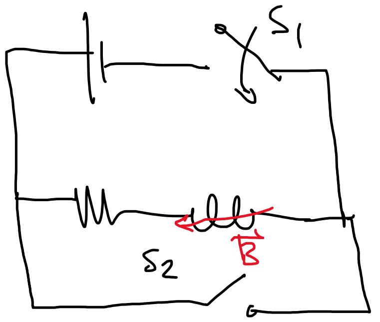


$$V_R = \varepsilon\left(1 - e^{-\frac{R}{L}t}\right)$$

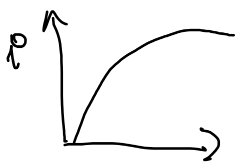
$$V_L = \varepsilon\, e^{-\frac{R}{L}t}$$

$$V_R + V_L = \varepsilon$$

RL switch off

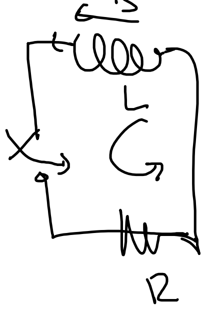


S_1 ON ; $\vec{B}$ inside L



energy is stored in L

S_1 OFF ; S_2 ON $\Rightarrow$ $\vec{B}$; energy will be decreasing to ze

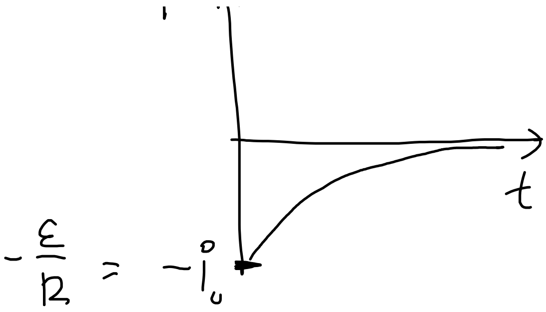


KVL: $V_L + V_R = 0$

$$L\frac{di}{dt} = -iR \Rightarrow \frac{di}{i} = -\frac{R}{L}t$$

$$\ln(\ldots\ldots)$$

$$i(t) = -i(0)\, e^{-\frac{R}{L}t}$$

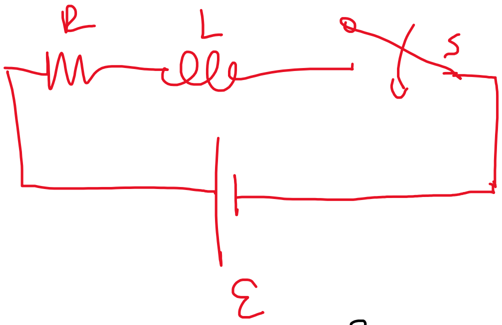


$$\rightarrow V_L = L\frac{di}{dt} = L\{-i(0)\}\left(-\frac{R}{L}\right)e^{-}$$

$$V_L = i(0)\,R\,e^{-\frac{R}{L}}$$

$$+\quad V_R = iR = -iRe^{-}$$

0



$R = 175\Omega$

$\rightarrow$ Current shouldn't be more than 36

to avoid damage.

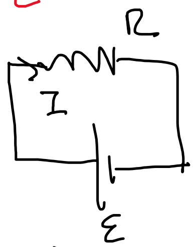
$\rightarrow$ current should not be more th

4.9 mA in the firs 58

$\varepsilon = ?$ $L = ?$ $\tau = ?$

$t \gg \tau$

$I \leq 36\,mA$



$$\varepsilon = iR = (36\,mA)(175\Omega) = 6.3\,V$$

$$i(t = 58\,\mu s) \leq 4.9\,mA$$

$$\underbrace{\frac{\varepsilon}{R}}_{36\,mA}\left(1 - e^{\frac{R}{L}t}\right) = 4.9\,mA$$

$$36\,mA\left(\qquad\right) = 4.9\,mA$$

$$1 - e^{-\frac{175}{L}(58\,\mu s)} = \frac{4.9}{36}$$

$$\underbrace{\ln\left(1 - \frac{4.9}{36}\right)} = \ln\left(e^{-\frac{175\times 5}{}}\right)$$

$$\ln(0.864) = -175\times$$

$$\sim -0.15$$

$$L = \frac{-175\times 58\times 10^{-6}}{-0.15}$$

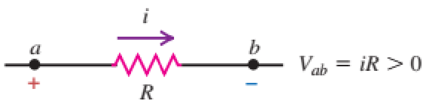
$$= 67667\times 10^{-6}\,H$$

$$L = 67.7\,mH$$

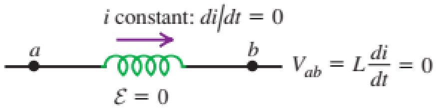
$$\tau = \frac{L}{R} = \frac{67.7\,mH}{175\,\Omega} = 0.387\,ms = 387\,\mu s$$

30.6 (a) The potential difference across a resistor depends on the current. (b), (c), (d) The potential difference across an inductor depends on the rate of change of the current.

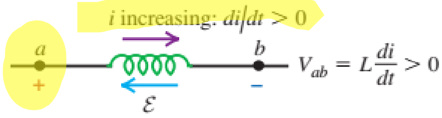
(a) Resistor with current i flowing from a to b: potential drops from a to b.



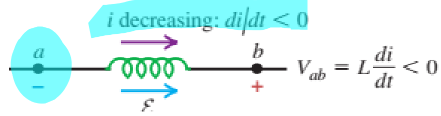
(b) Inductor with *constant* current i flowing from a to b: no potential difference.



(c) Inductor with *increasing* current i flowing from a to b: potential drops from a to b.



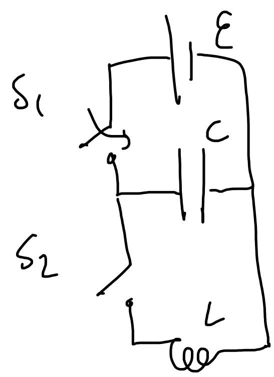
(d) Inductor with *decreasing* current i flowing from a to b: potential increases from a to b.



Inductors REACT to Current CHANGE!

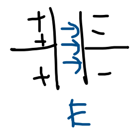
inductors make "smooth" current changes

LC circuits



S_1 ON ; S_1 OFF

(1) ⇒



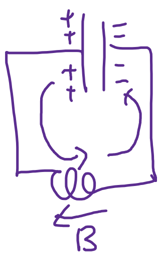
charged

$\vec{E}$ is stored in a charged capacitor.

(2) S_2 ON   LC circuit

(2) charged capacitor is creating a current towards L

not constant



L will react to current change $\vec{B}$ will be formed.

Kirchhoff law $V_C + V_L = 0$

$$\frac{q}{C} + L\frac{di}{dt} = 0$$

$$i = \frac{dq}{dt} \qquad \frac{di}{dt} = \frac{d^2q}{dt^2}$$

$$\frac{q}{C} + L\frac{d^2q}{dt^2} = 0$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC} q = 0 \Rightarrow \boxed{\frac{d^2q}{dt^2} = -\frac{1}{LC} q}$$

$q(t) = \sin\ ;\ \cos$

unit analysis

$$\left[\frac{C}{s^2} = \frac{1}{HF} C\right] \Rightarrow [HF = s^2]\ ;\ \boxed{HF = s}$$

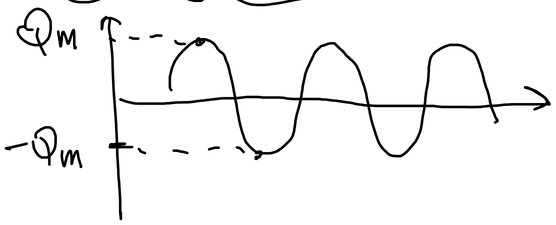
$$\frac{1}{\sqrt{LC}} = \frac{1}{[sec]}$$

$q = A \sin(\omega t)$ — ω: ang. freq.

$$\frac{dq}{dt} = A\omega \cos(\omega t)$$

$$\frac{d^2q}{dt^2} = -A\omega^2 \sin \omega t = -\omega^2 \underbrace{A \sin(\omega t)}_{q}$$

$$\ddot{q} = -\omega^2 q$$



$q = A \sin \omega t$

frequency of the oscillation $\omega = \frac{1}{\sqrt{LC}}$

$A = Q_{max}$

ω = angular freq $= 2\pi f$

f = frequency $\quad f = \frac{1}{2\pi\sqrt{LC}}$

$$\boxed{q(t) = Q_{max} \sin\left(\frac{1}{\sqrt{LC}} t\right)}$$

→ charge of capacitor

$$V_C = \frac{q}{C} = \frac{Q_{max}}{C} \sin\left(\frac{t}{\sqrt{LC}}\right) = \frac{\varepsilon d}{d} \sin(\omega t) = \varepsilon \sin \omega$$

$V_C + V_L = 0 \qquad V_L = -\frac{Q}{C} \sin\left(\frac{t}{\sqrt{LC}}\right) = -\varepsilon$

$V_L = L \frac{di}{dt}$, $\quad \frac{di}{dt} \sim \frac{d^2q}{dt^2}$

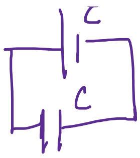
$i = \frac{dq}{dt}$

$V_L = L \frac{d^2q}{dt^2}$ — same

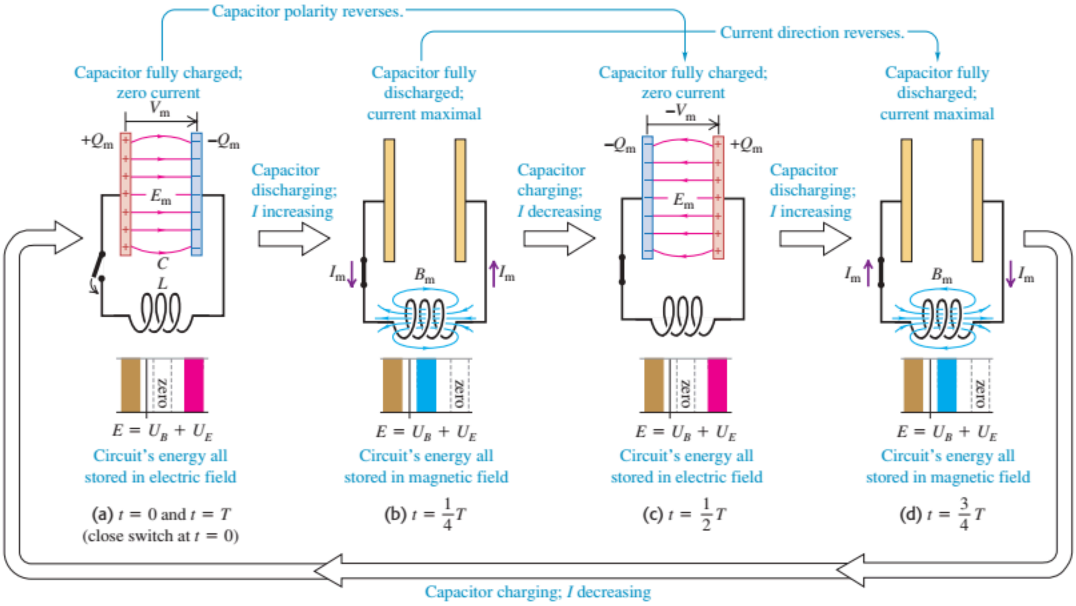
$V_L = L\, Q_{max} (-1) \sin(-$

c

$q_{max} = \mathcal{E}C$



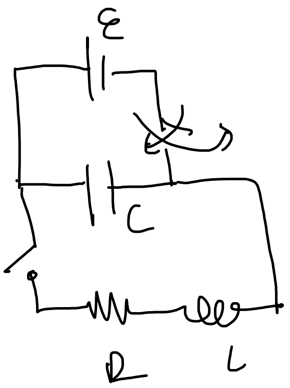
$\mathcal{E}C$



30.14 In an oscillating *L-C* circuit, the charge on the capacitor and the current through the inductor both vary sinusoidally with time. Energy is transferred between magnetic energy in the inductor (U_B) and electric energy in the capacitor (U_E). As in simple harmonic motion, the total energy E remains constant. (Compare Fig. 14.14 in Section 14.3.)

1 full cycle $\quad$ freq $= \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$

RCL $\quad$ RLC circuit.



charged cap with ε is connect to R & L

KVL $\quad V_C + V_L + V_R = 0$ $\qquad i = \frac{dq}{dt}$

$$\frac{q}{C} + L\frac{di}{dt} + iR = 0$$

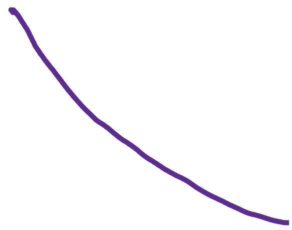
$$\frac{q}{C} + L\frac{d^2q}{dt^2} + R\frac{dq}{dt} = 0$$

$$\left\{ \frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{1}{LC}q = 0 \right\}$$

$$\longrightarrow \quad q(t) = A\cos(\omega' t)\, e^{-\frac{R}{2L}t}$$

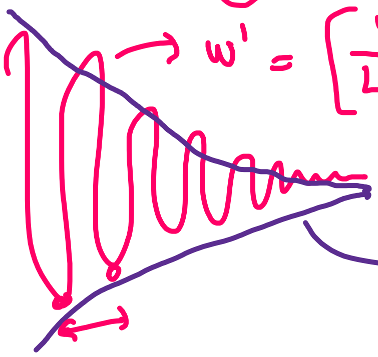
$\left(\ldots \sqrt{1 \quad R^2} \right\}$ ang.

$\omega = \sqrt{\frac{1}{LC} - \frac{\ }{4L^2}}$ Re



$q(t) = A \cos \omega' t \; e^{-\frac{R}{2L}t}$

$q(t) =$



$\omega' = \left[\frac{1}{LC} - \frac{R^2}{4L^2}\right]^{1/2}$

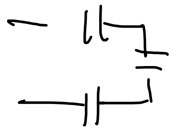
$f = \frac{1}{T}$

$e^{-\frac{R}{2L}t}$

FINAL

No GAUSS !!

No C circuits



Kirchoff ✓, RC, RL, RLC - - -

$\vec{B}$!!